Autodesk Vault 2009

Managing Your Data



January, 2008

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Data Management

Learn about the Autodesk® data management system and its configurations.

Key Terms

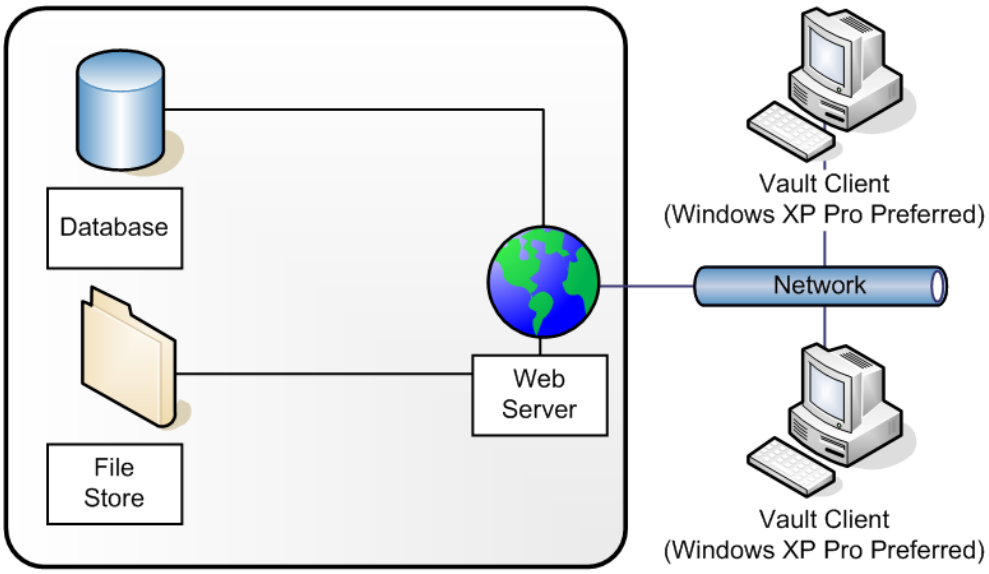
Term	Definition
Autodesk® data management server	The server used in conjunction with Autodesk® Vault, Autodesk® Productstream®, and Content Center for storing files and version data.
Autodesk Vault	A system for data management and version control. A general purpose application for interacting with a vault on the Autodesk data management server. You can browse the complete vault structure, add any file to the vault, and perform most other file-based operations, depending on your level of permission.
client	Applications that access the Autodesk data management server. These include stand alone applications such as Autodesk Productstream as well as integrated application add-ins such as Autodesk Vault for Inventor or Autodesk Vault for Microsoft® Office.

Term	Definition
data management	A means to organize and track files and design modifications through the design process.
server	The central location where all the vault data is stored and from which it is retrieved. The server must always be online for clients to access the vault. The vault server is configured using ADMS Console.
vault	The logical combination of a vault database and a vault file store that contains all of the information managed by Autodesk Vault. Each installation of Autodesk Vault can maintain several independent vaults. A single client can connect to several different vaults, but not simultaneously.

About Autodesk Vault

Autodesk Vault is a workgroup data management system for sharing design data across a project team. The vault is a file management and version control system for all engineering and related data providing design team members with a central and secure collaborative environment. Autodesk Vault consists of two components: the Autodesk data management server and vault clients. The server stores the master data files of all design information. The clients provide access to the files stored on the server.

Design teams use Autodesk Vault for version control and to store and share all types of engineering files and related data. Files can be Autodesk Vault, AutoCAD®, Autodesk® DWF™ (Design Web Format), FEA, CAM, Microsoft® Word, Excel®, or any other file used in the design process.



Single Vault Server (Windows 2003 Server Preferred)

Autodesk Vault

Autodesk Vault is a general purpose application for interacting with a vault on the Autodesk data management server. You can browse the complete vault structure, add any file to the vault, and perform most other file-based operations, depending on your level of permission.

Autodesk Vault Add-ins for Design Applications

Add-in clients provide basic vault functions within the environment of a parent application. Add-ins maintain application-specific data relationships when adding files to a vault.

There are add-ins available for Autodesk and non-Autodesk® design applications.

NOTE As a general rule, if an integrated client is available for a particular application, managing files using that client minimizes loss of data, such as the assembly relationships. We recommend that you use integrated clients whenever possible.

Microsoft Office Add-in

The Microsoft® Office Add-in performs basic vault functions on documents, spreadsheets, and other non-CAD data within any of these Microsoft Office applications: Word, Excel, and PowerPoint®.

Data Management

All versions of files checked into the vault are retained, along with any file dependencies, providing a living history of the project. Team members have access to the files and data stored on the server, and the history of files. They check out files to prevent more than one member from editing the same file at the same time. After a file is checked back into the vault, team members can refresh their local copies.

Vault Clients 2

Learn how to configure vault clients and how to log in to a vault.

About Vault Clients

Vault clients include the following software applications:

Client Software	Description
Autodesk® Vault	A stand-alone client providing full access to a vault.
Vault add-in for Autodesk® Inventor™	Manage parts, assemblies, and other Autodesk Inventor files that are stored in the vault from the Autodesk Inventor interface.
Vault add-in for AutoCAD®	Perform basic vault functions within AutoCAD, AutoCAD® Mechanical, AutoCAD® Electrical, and Autodesk® Mechanical Desktop®.
Vault add-in for Microsoft® Office	Perform basic vault functions on documents, spreadsheets, and other non-CAD data within any of the Microsoft Office applications.

Logging In to Vault

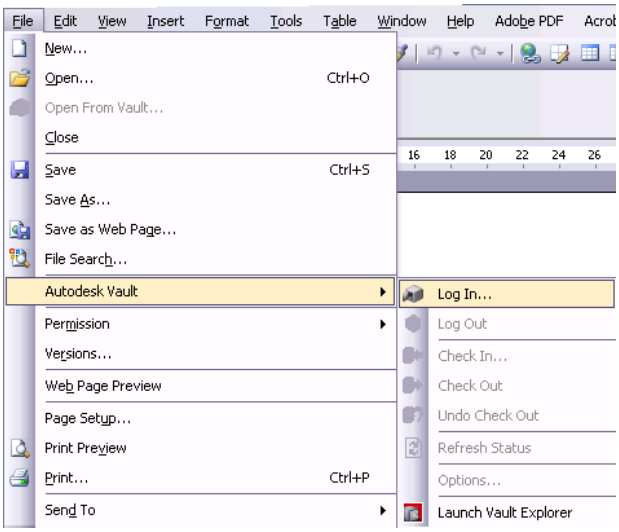
When Autodesk data management server is installed, a vault and a user account are created. The name of the vault is Vault. The account name is administrator. No password is assigned to the account and the name is not case sensitive. The administrator account has full administrator privileges.

NOTE If a user account has already been created for you, log in using your user name and password.

Log into Autodesk data management server the first time

1 Use one of these methods:

- On the Start menu, click Programs ➤ Autodesk ➤ Autodesk Data Management ➤ Autodesk Vault (version).
- On the Start menu, click Programs ➤ Autodesk ➤ Autodesk Data Management ➤ Autodesk® Productstream® (version).
- In an application with an integrated add-in, click File ➤ Autodesk Data Management Server ➤ Log In.



2 In the Log In dialog box, verify the following:

User name The name for the vault account.

Password	The password associated with the vault account.
Server	The name of the computer on which the vault server is installed.
Database	The name of a vault database located on the specified server. The default is "Vault". Click the browse button to select from a list of available databases on the server.

3 A default user ID, vault server, and database are set up for you to get started using the vault right away. If a user account has already been created for you, use your own account information. If a user account has not been created for you, or the default values are not present, contact your vault administrator.

4 Use either the default values or, if an account has been created for you, use your own account information. The default values are:

5 `User Name:` Administrator

`Password:` empty

`Server:` Name of the computer on which the vault server is installed. Use localhost if the server is installed on the same machine as the client.

`Database:` Vault

6 Click OK.

You are logged into the vault.

NOTE You have the option to save the account information to automatically log into the vault the next time.

Autodesk Vault 3

Manage files using Autodesk® Vault, the general purpose interface to Autodesk® data management server.

Key Terms

Term	Definition
check in	Returning a modified file to the vault. The previous version is retained in the file history.
check out	Retrieving a read/write copy of a file stored in the vault for editing.
latest version	The most recent version of a file stored in the vault.
local copy	The copy of the vault source file placed in the local working directory.
previous version	An older version of a file stored in the vault. Previous versions of a file are retained and are accessible at any time.

Working with Autodesk Vault

Use Autodesk Vault to manage folders and files within a vault.

- Start Vault
- Manage folders

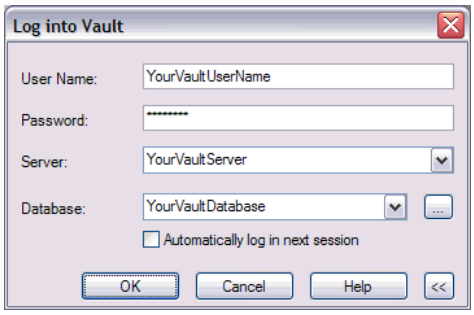
- Manage files
- Manage file versions
- Perform searches
- Package files

Starting Autodesk Vault

Start Vault

- On the Microsoft® Windows® Start menu, click Programs ➤ Autodesk ➤ Autodesk Data Management ➤ Autodesk Vault (version).
- On the desktop, double-click the Autodesk Vault icon.

1. In the Log in dialog box, enter your user name, password, the server name, and the name of the computer.
2. Click OK.



NOTE You may save your account information to automatically log into the vault upon next use.

Managing Vault Folders

Using Vault, create the folder structure for the vault.

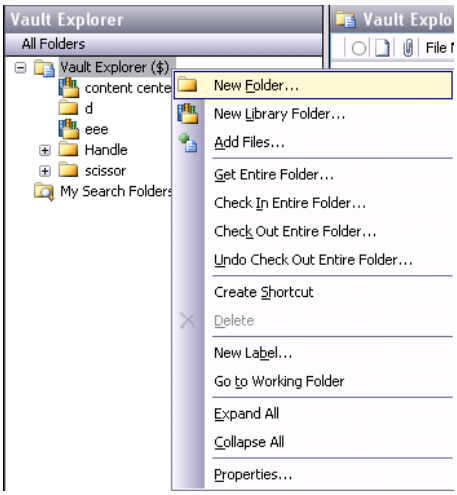
Creating the Folder Structure

Autodesk Vault is used to create the folder structure for the vault. The root directory in a vault is represented as *$/*.

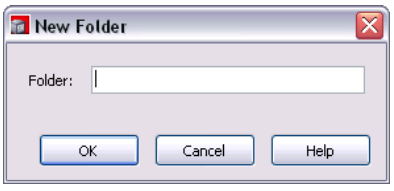
NOTE Autodesk® Inventor™ only. Because Autodesk Vault uses the Autodesk Vault project file as a reference for file organization, set the folder structure inside the vault identical to the folder structure of the project on the local computer.

Create a folder

1 Right-click the Vault (*$*) folder, and then select New Folder.



2 In the New Folder dialog box, specify a name for the folder.

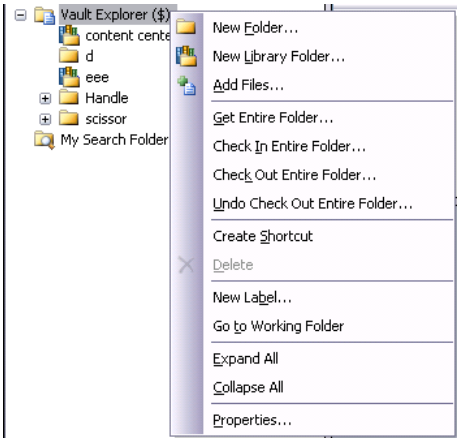


3 Click OK.

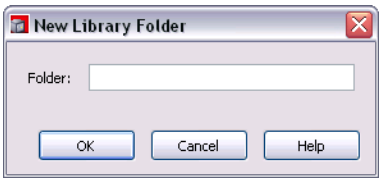
The folder is created below the root level of the vault. Additional subfolders might be created automatically when you add files to the vault using the Vault add-in for Autodesk Vault and for AutoCAD®-based products.

Creating Library Folders

1 To add a library folder, right-click the root level directory ($), and then click New Library Folder.



2 In the New Library Folder dialog box, enter a name.



You can create library folders directly from the root level of the vault or below another library folder.

Managing Vault Files

Manage design and project files using Vault.

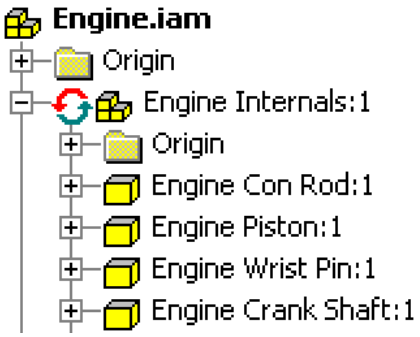
Any non CAD files can be added to a vault using Vault. When adding CAD data from Autodesk Vault, or any of the AutoCAD-based products, be sure to add the files from within the CAD application to maintain file relationships.

- For Autodesk Vault design files, use the Autodesk Vault add-in for Inventor.

- For DWG host files containing xrefs in applications such as AutoCAD, AutoCAD® Mechanical, AutoCAD® Electrical, or Autodesk® Mechanical Desktop®, use the Autodesk Vault add-in for AutoCAD.

File relationships are maintained among files stored in the vault. When getting a file, checking a file out, or checking a file in, the files related to a selected file can be included. The following terms refer to related files:

Children	Files on which the selected file depends.
Parent	Files that are dependent on the selected file.



Autodesk Vault can be used to manage and track the many different types of files and data associated with a design project, including Autodesk Vault and AutoCAD data, to documents and spreadsheets.

File Status Icons

The following icons indicate the status of files in the Autodesk Vault window, the Vault browser in Autodesk Vault, , and the Xrefs Manager in AutoCAD-based products.

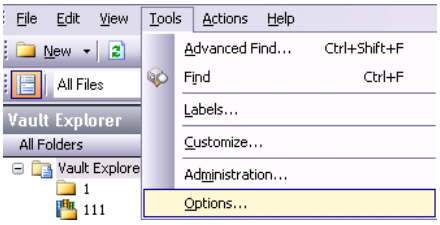
Icon	Description
	Not in the vault or the file has missing parent/child relationships. Use the Add Files command to add the item to the vault or resolve the missing links.
	Checked in to the vault and the version you are working on is the same as in the vault. Also referred to as the Latest Version.

Icon	Description
	Checked in to the vault, but the version you are working on is newer than the latest version in the vault. This typically means that the local file was changed without checking it out. To save the changes, check the file out, and make sure the Don't Get Local Copy option is selected.
	Checked in to the vault, but the version you are working on is older than the latest version in the vault. This typically means that another user made changes since your last update. Use Get Latest Version to update to the latest available version.
	Checked out to you and the version you are working on is the same as in the vault. Also referred to as the Latest Version.
	Checked out to you, but the version you are working on is newer than the latest version in the vault. This typically means that you made changes to the model since the last time you checked out the file, but have not checked it back in.
	Checked out to you, but the version you are working on is older than the latest version in the vault. This typically means that you started with a version from the vault that was older than the latest, and checked it out to promote it to the latest.
	Checked out to another user, and the version you are working on is the same as in the vault. Also referred to as the Latest Version. This typically happens if the other user did not check changes back into the vault. Check with the other user before proceeding.
	Checked out to another user, but the version you are working on is newer than the latest version in the vault. This typically happens if the user checked in saved changes to the vault, but kept the file checked out. Use Get Latest Version to update to the latest available version.
	Checked out to another user, but the version you are working on is older than the latest version in the vault, and another user checked out this file. Use Get Latest Version to update to the latest available version.

Icon	Description
	Autodesk Vault only: User is not logged into the vault.
	Has attachments. Expand the tree to see what files are attached.
	Has third-party OLE linked files. This is displayed in the Autodesk Vault browser only. Expand the tree to see what files are linked.
	An an error occurred when adding your project to a vault. You must resolve the errors to add the files to the vault. This icon can also mean that you need to set a project folder mapping in the Map Folders dialog box.
	File doesn't exist in the expected working folder. This is common in two cases: 1) you created a new file but have not saved it to disk yet, and 2) an attachment isn't in the same folder as the file it is attached to. This is a requirement for files that are attached to Autodesk Vault documents.
	Library folder. The files within this folder are not intended to be edited when used in context of another assembly.

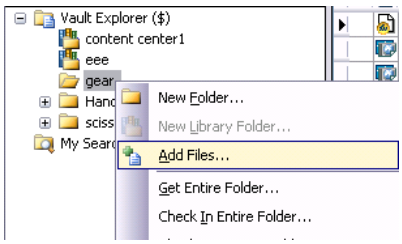
Choose between the large and small icons

1. Select Tools ➤ Options in Vault.
2. By default, the small icons are used to indicate status. Turn on the Use large status icons check box to display large icons.



Adding Files to a Vault

1. Right-click a vault folder, and then click Add Files.



2. In the Add files dialog box, select the file to add to the vault.
3. Click OK.

After adding a file to the vault, you can delete the local copy. When you want to view or modify a copy of the file, use Autodesk Vault to retrieve a version of the file and check it out.

Moving Files within Vaults

1. Ask all users to check in their local data and delete the local copies.
2. Using Autodesk Vault, drag one or more files from the existing location to a new location, to reflect the new file organization.
3. Drop the file in the new location and select Move from the context menu.
4. Check out the Autodesk Vault files and resolve any broken links.
5. After the vault files are moved and broken links are resolved, ask all users to log into the vault and get the latest version of the top-most file to their working folders.

Attaching Files in the Vault

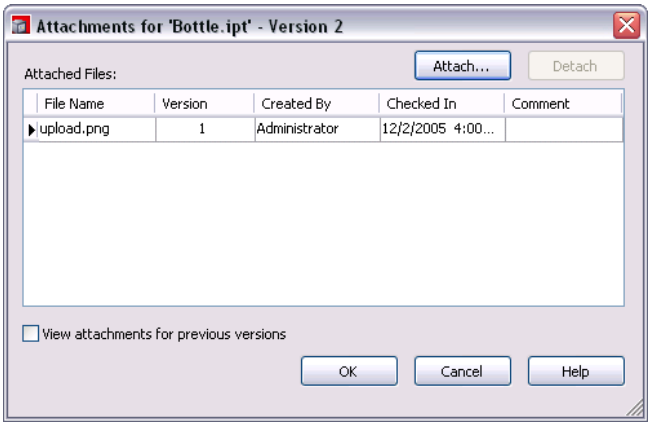
An attachment among files links them in the vault. You can manually add attachments among any two or more files within the vault. Using attachments, you can check out all the files that are linked, and work on them as a unit.

You can attach different file types. For example, you can attach a set of images to a Microsoft® Office Word document, and you can attach assembly instructions to the actual CAD assembly files.

NOTE Do not attach files that can be assembled (Autodesk Vault) or referenced together (AutoCAD-based products).

Attach a file

1. Select a file in the vault for which you want to define an attachment.
2. In the Vault menu, click Attachment or right-click the file to which you want to add an attachment, then click Attachment.
3. The Attachments dialog box displays the current attachments.
4. Click Attach.



5. A file browser is displayed. Browse to and highlight the files you wish to attach. To select multiple files, use a crossing window or hold down CTRL or SHIFT.
6. Click Open.

   The selected files are listed in the Attached File list in the Attachments dialog box.
7. When you finish all attachments, click OK.

Remove an attachment

1. Select the file in the vault from which to remove an attachment.

2 On the Vault menu, click Attachments.

3 In the Attachments dialog box, select the file to remove.

4 Click Detach.

5 Click OK to detach the selected files and close the Attachments dialog box. A new version of the file is created in the vault with the selected files removed from it.

As a project grows, it may become necessary to move files and change the folder structure within a vault. Use Autodesk Vault to reorganize files and folders on the server.

NOTE Reorganizing files is only performed using Autodesk Vault. Never move files that are located in the working folder.

Renaming Files Within Vaults

As a project grows, it might become necessary to rename files within a vault. You must use Autodesk Vault to rename files so that file relationships are maintained.

If you want to rename the folder, make sure all the files in the folder are checked into the vault. A folder cannot be renamed if any of its files has an external dependency.

For more information about renaming vaulted files, see the Help within Autodesk Vault.

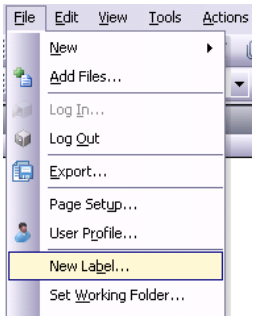
Labeling Files

In Vault, you can label development milestones such as customer proposals, design reviews, or concept variations so they can be easily identified. Using labels, you can establish baselines and track progress for work-in-progress files. You can also rename labels, extract labeled files from the vault using Pack and Go, and restore labeled files.

When you create a label, the most recent checked in version of every file in the project is assigned that label. There is no limit to the number of labels that can be assigned to a project. You cannot create more than one label with the same name. For more information on labels, see Labeling Files in the Autodesk Vault Help.

Label a file

1 Click File ➤ New Label.



2 In the New Label dialog box, enter a name for the label that indicates the milestone for that project. Label names can contain any alphanumeric text, excluding \ / : * ? " < > |.

3 Optionally, enter a comment to describe the label. Labels help identify the contents of the label.

4 Click OK.

Copying Designs

Create new products based on existing Autodesk Vault or AutoCAD designs by copying the entire assembly structure along with the related 2D drawings for 3D models within Autodesk Vault. You can manipulate existing assemblies to derive new designs and "one offs" complete with documentation.

In the Copy Design dialog box, you can selectively choose which parts of an existing design to copy, reuse, exclude, or replace. A naming scheme can be defined for the files being copied to the new design. A prefix and a suffix can optionally be added to the files automatically. The new files can also be automatically incremented if the original file names end in an integer. This is useful for when CAD files are named using item or part numbers. Autodesk Vault presentation files and drawing files can be automatically named to match the names of their direct part or assembly child.

For more information, see Copying Designs in the Autodesk Vault Help.

Copy a design

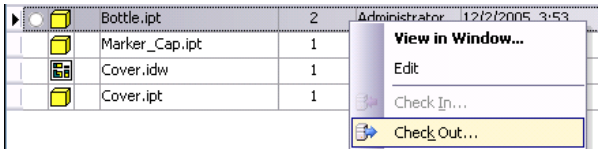
1. Select a file in either the main pane or the preview pane.
2. Click Copy Design in the toolbar.
3. In the Copy Design dialog, specify which parts to reuse, exclude, copy, or replace.
4. Click OK.

Understanding Check Out

When you check out a file, the read-only attribute of the local copy changes to read/write, and you can edit it. No one else can modify a file you check out until you check it back into the vault, but they can retrieve read-only copies. Only one member of a team can check out a file at one time.

Check out a file to edit

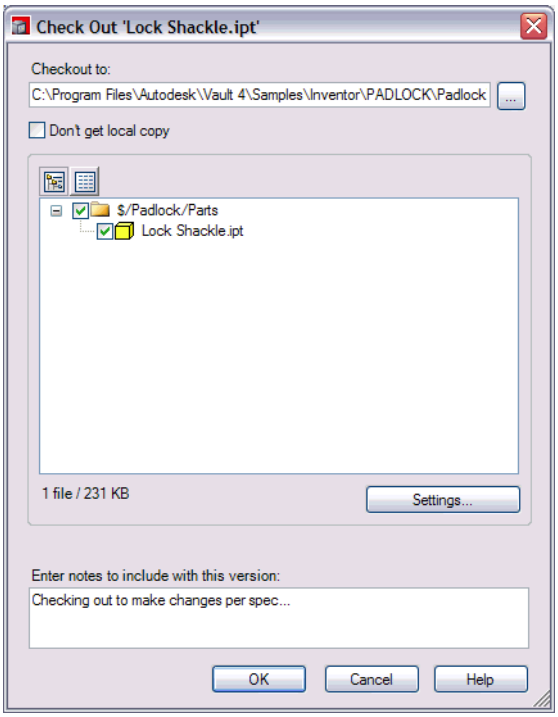
1. In Autodesk Vault, right-click a file, and then click Check Out.



2. In the Check Out dialog box, you can add a comment.

NOTE Autodesk Vault relies on consistent file locations to resolve links. Check files out to the default working directory.

As you check files out and in, use comments to add information for managing versions and to track the actions performed during the last working session. Use the advanced options to check out the parents and children of the selected file.



3 Click OK.

A check mark next to a file indicates that the file is currently checked out for editing. No other team members can check out a file that is currently checked out. However, others can get read-only previous versions and latest checked in versions of files that are currently checked out.

NOTE You can also right-click a folder, and click Check Out Entire Folder.

Understanding Check In

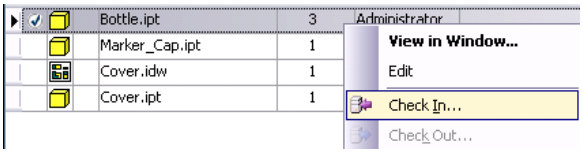
Check In returns the local copy of the file to the vault with the changes that have been made. When a file is checked in, it becomes the latest version and is accessible to others.

You must check in files from the working folder from which they were checked out. If files are moved from the working directory, you cannot check them back into the vault but you can undo the check out. The file properties

CheckOutMachine and CheckOutLocalSpec specify the required location of the file.

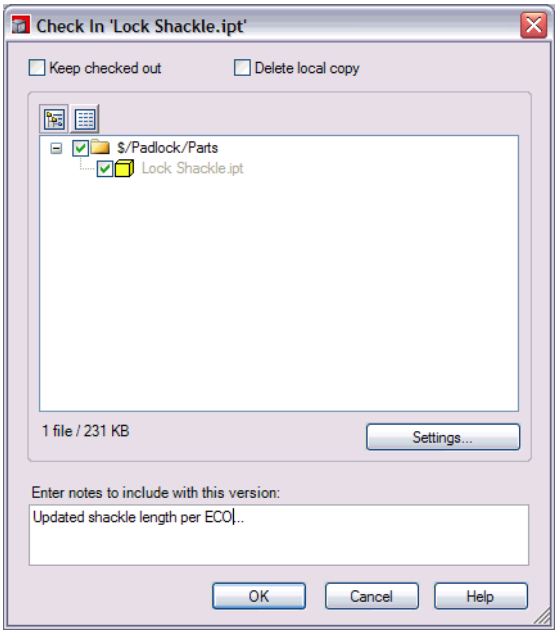
Check in a file

1 In Vault, right-click on a file, and click Check In.



2 In the Check In dialog box, you can:

- Check in the file with your changes.
- Check in your changes and keep the file checked out for further editing.
- Delete the local copy upon check in if you check in the file.
- Add a comment containing information about changes made in the file.
- Use the Settings options to check in the parents and children of the selected file.



3 Click OK.

The information for the file in Vault indicates that the file is checked in. The file becomes the latest version and other team members can check it out.

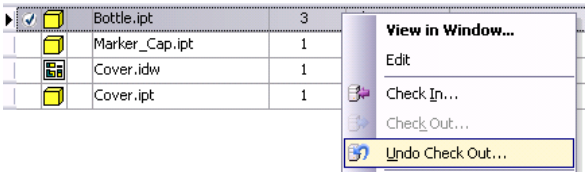
NOTE You can right-click a folder, click Check in Entire Folder.

Understanding Undo Checkout

Undo Checkout removes the checked out status on the file. The latest version of the file that was checked in is restored to the vault.

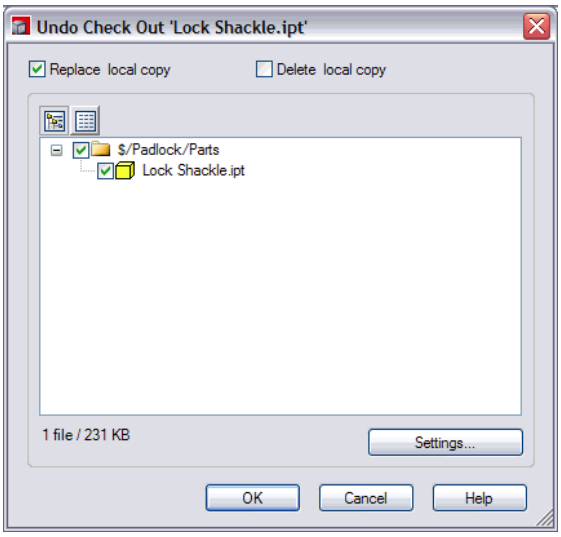
Undo a Checkout

1 In Vault, right-click a file, and then select Undo Check Out.



A prompt indicates that undoing a checked out file results in a loss of changes.

2 To make the local file the same as the file in the vault, select Replace working copy.



3 Click Settings to specify whether or not the children of the selected file are included.

4 Click OK.

The file has been returned to the vault. Any changes made to the local copy are lost.

NOTE You can right-click on a folder and then select Undo Check Out Entire Folder.

Displaying Version History

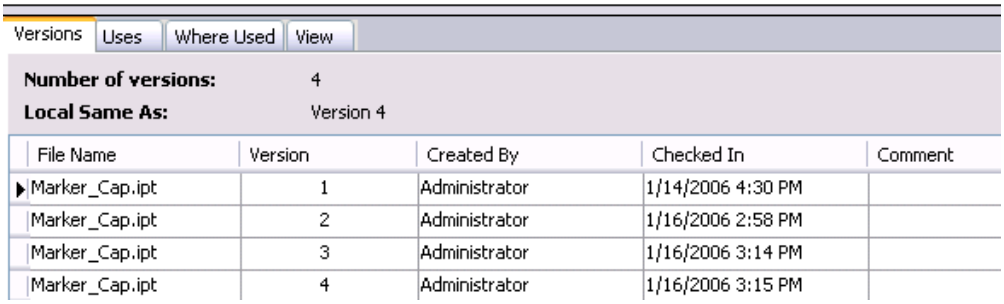
In addition to tracking versions, Autodesk Vault tracks properties associated with the data. You can use the version information and properties to help manage and search for data.

The files stored in the current vault are listed in the main pane of the Vault. Version and file information for the selected file is displayed in the bottom pane under four tabs; Versions, Uses, Where Used, and View. Each heading at the top of the pane represents a property field for the file. As you add more properties to a file, more fields are available for customizing the views within the tabs.

1. Select a file in Autodesk Vault.
2. Click the Version tab in the preview pane.



File Name	Version	Created By	Checked In	Comment
Marker_Cap.ipt	4	Administrator	1/16/2006 3:15 PM	
Marker_Handle.ipt	2	Administrator		
Market_Insert.ipt	2	Administrator		



Versions | Uses | Where Used | View

Number of versions: 4

Local Same As: Version 4

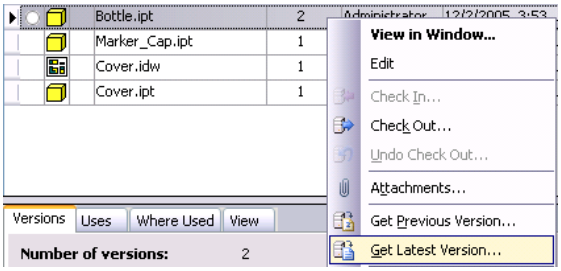
File Name	Version	Created By	Checked In	Comment
Marker_Cap.ipt	1	Administrator	1/14/2006 4:30 PM	
Marker_Cap.ipt	2	Administrator	1/16/2006 2:58 PM	
Marker_Cap.ipt	3	Administrator	1/16/2006 3:14 PM	
Marker_Cap.ipt	4	Administrator	1/16/2006 3:15 PM	

Getting Latest Versions

Get Latest Version retrieves a read-only copy of the most recent design data that is checked in. You cannot modify it until you check it out using Autodesk Vault or a Vault add-in product.

Get the latest version of a file or folder

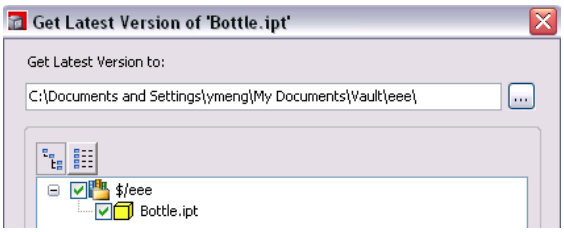
1 In the Vault window, right-click the file or folder, and then click Get Latest Version.



2 In the Get Latest Version dialog box, click OK.

The most recent version is copied to the local working folder. If the most recent version in the vault is different from the copy currently in the working folder, you are prompted about overwriting data.

Click Settings to get the parents and children of the selected file.



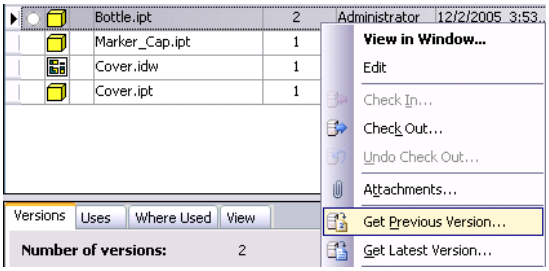
NOTE Autodesk Vault only: Library files and referenced files outside of the workspace are retrieved to a directory parallel to the working folder. To ensure that the correct version of the libraries are referenced, check out the project file from the vault and modify the library search paths to point to the new location.

Getting Previous Versions

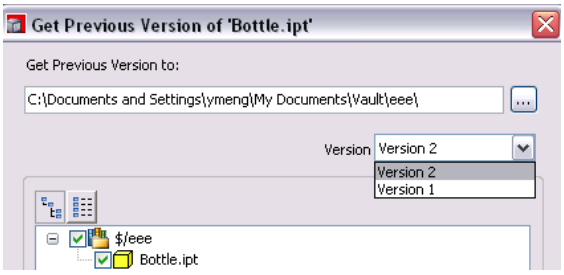
Get Previous Version retrieves a past version of a file or a project and places a read-only copy in your working folder. Historical versions can never be modified. You can only create a new version of a file.

Get a previous version of a file or project

1 In the Vault window, right-click a file or project, and then click Get Previous Version.



2 In the Get Previous Version dialog box, select a version of the file or project to retrieve.



3 If the file has parents and children to get, click Settings, and then specify which related files are retrieved as well.

4 Click OK.

A read-only copy of the file is placed in the local working folder. You can view the file but you cannot modify it. To modify the file, you must check it out.

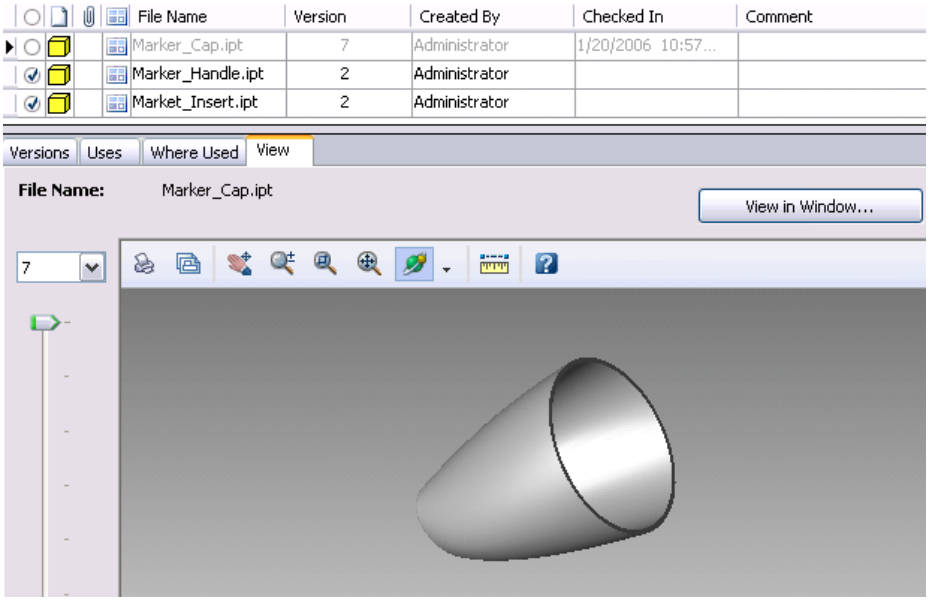
NOTE Autodesk Vault only: Library files and referenced files outside of the workspace are retrieved to a directory parallel to the working folder. To ensure that the correct version of the libraries is referenced, check out the project file from the vault and modify the library search paths to point to the new location.

Viewing Previous Versions

1 Select a file in Autodesk Vault.

2 Click View tab in the preview pane. The associated *.dwf* file of the selected file will be displayed using Autodesk DWF Viewer.

3 Choose the file version which you want to view by the version slider in the View tab.

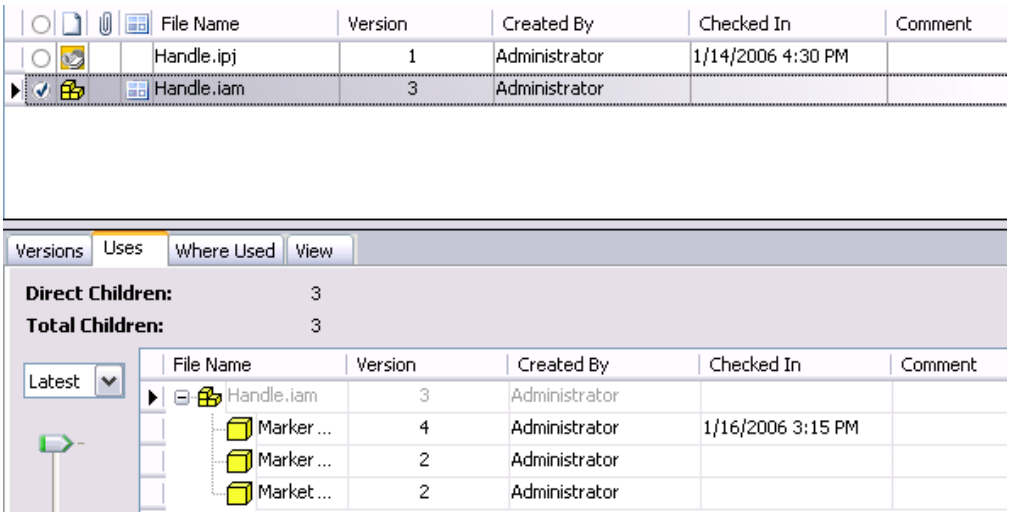


Displaying Other Files Used

1 Select a file in Autodesk Vault.

2 Click the Uses tab in the preview pane.

The selected model is shown at the top. Other files used by the currently selected file, and the children and attached files are listed in a hierarchy as shown in the following illustration.

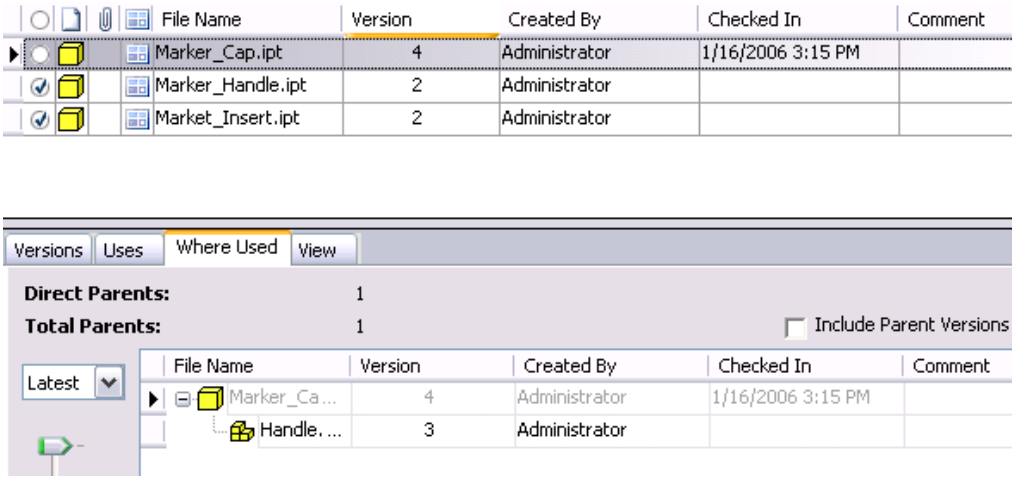


Displaying Where Files Are Used

1 Select a file in Autodesk Vault.

2 Click the Where Used tab in the preview pane.

A list of parents in which the selected file is used is displayed. The component is listed at the top. Each assembly using the part is listed as follows.



Publishing DWF Files

DWF files can be automatically published for CAD files checked in to the vault using the vault add-ins for supported CAD applications. The *.dwf* attachments are only created for files that have changed or for files that do not have *.dwf* files published already. Automatic *.dwf* publishing can be turned off to save room and keep the size of the vault down. You can also specify a default folder location for the published *.dwf* files. For more information on automatic *.dwf* publishing, see Help in Autodesk Vault or each of the supported CAD application add-ins.

Performing Searches

Use the Advanced Find dialog box for basic searches and to create customized search criteria for locating specific information in your data. If the Content Indexing Service is enabled on the Autodesk data management server, you can search the content of files as well. For more information on searching, see Autodesk Vault Help. For more information on enabling content indexing, see ADMS Console Help.

Performing Basic Searches

1. On the Tools menu, select Advanced Find, and then click the Basic tab.
2. Enter your search criteria. Click Find Now.

   The results of the search are displayed at the bottom of the Advanced Search dialog box.

Performing Advanced Searches

1. On the Tools menu, select Advanced Find, and then click the Advanced tab.
2. In the Property list, select a document property.

   The Property list contains all the properties associated with the files stored in the current vault.
3. In the Condition list, specify a condition by which to search.

4 Enter a value to be used in the search criteria.

5 To add the search criteria to the list of stored searches, click Add.

6 Click Find Now.

The results of the search are displayed at the bottom of the Advanced Search dialog box.

Packaging Files with Pack and Go

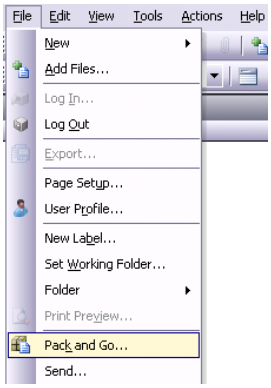
Pack and Go packages a file and all of its referenced files in a single location. All files that are referenced by a selected file are included in the package unless otherwise specified in the Pack and Go dialog box.

Use Pack and Go to:

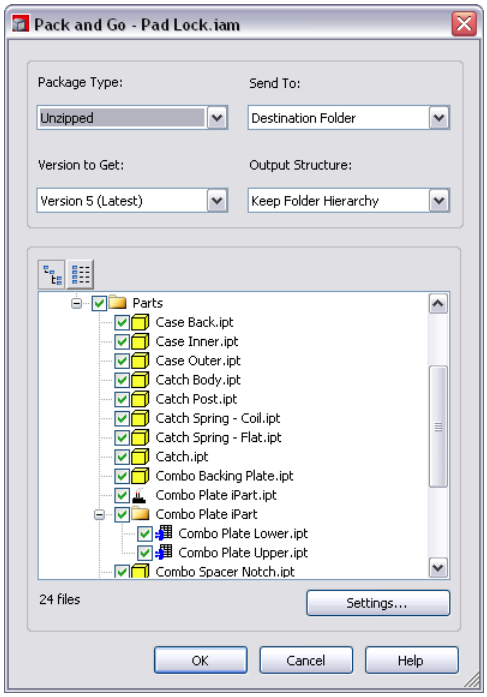
- Archive a file structure.
- Copy a complete set of files, while retaining links to referenced files.
- Isolate a group of files for design experimentation.
- Retrieve a previous version of a design and copy it into an isolated folder.
- Send the package to a mail recipient.

Package a set of files

1 Select a file from the file list in Autodesk Vault. Click File ➤ Pack and Go.



The Pack and Go dialog box shows a list of the files to be packaged.



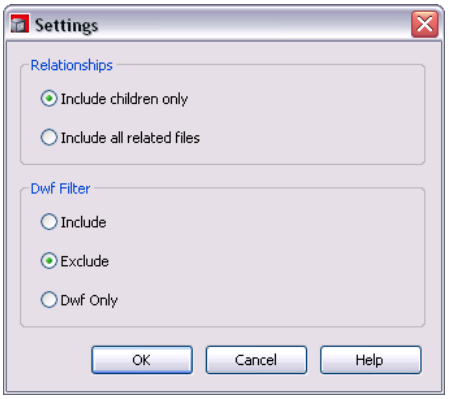
2 You can:

- Package the files to a *.zip* file or create a DWF package, a single *.dwf* file with each file represented as a page in the *.dwf* file.
- Select which version of the files to package.
- Specify the destination for the package, either a folder on the local computer or a mail recipient.
- Determine the structure of the files being packed. You can flatten all files and place them in a single folder, or preserve the structure as it appears in the vault.

NOTE If you choose to flatten all files and place them in a single folder, any file sharing the same name as another file is automatically renamed to avoid file name collisions. This may require that you manually resolve the renamed files in their native CAD systems.

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3 Click Settings to select which related files will be included in the package and whether or not *.dwf* files are included.



4 Click OK . If the destination for the package is a mail recipient, an empty e-mail window is displayed with e package included as an attachment.

For more information on packaging files, see Autodesk Vault Help.

Customizing Vault

In Vault, you can customize your work environment. By suppressing dialog boxes for certain commands and controlling the default responses to various prompts, as well as when prompts appear you can greatly streamline your workflow. For more information on customizing your workflow and suppressing recurring prompts, see Autodesk Vault Help.

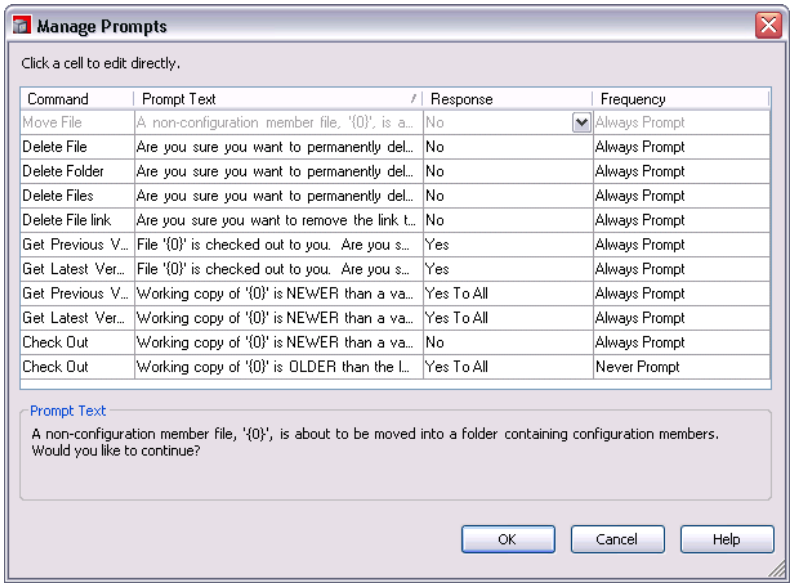
Customize your work environment

1 Click Tools ➤ Options. In the Options dialog you can do the following:

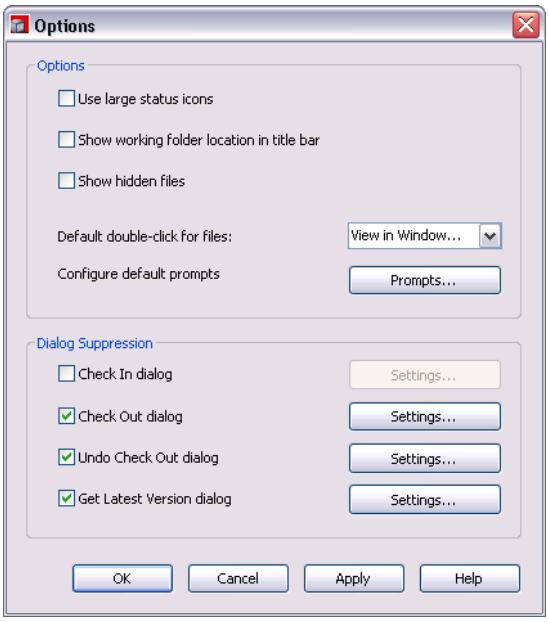
- Select Use large status icons to display large icons.
- Select Show working folder location in title bar to display the location of the working folder in the user interface.
- Select Show hidden files to display the hidden files in the file list such as *.dwf* files.
- When you double-click a file in the vault, the default action is to display the read-only file in a new window. If Edit is selected as the

default, the file is checked out of the vault and opened in the associated application for editing. To change the default action, select an action from the drop-down list.

- Configure prompts. To do this Click Prompts... A list of prompts appears. To modify the default response to a prompt click the dropdown menu in the Response cell. Go to the Frequency cell to adjust when and how often a prompt appears. Click OK.



2 In the Dialog Suppression Panel, there are four dialog suppression options. To suppress a dialog, select the checkbox next to it. Click the settings button and adjust the default settings. Click OK.


Options
Options
Use large status icons
Show working folder location in title bar
Show hidden files
Default double-click for files:
View in Window...
Configure default prompts
Prompts...
Dialog Suppression
Check In dialog
Settings...
Check Out dialog
Settings...
Undo Check Out dialog
Settings...
Get Latest Version dialog
Settings...
OK
Cancel
Apply
Help

Working with Autodesk Inventor

4

Learn how the Autodesk® Vault add-in integrates with Autodesk® Inventor™. Before you proceed, make sure you are familiar with setting up Autodesk Inventor projects and the structure of your design project.

Key Terms

Term	Definition
Vault add-in for Autodesk Inventor	The Autodesk Vault client interface that is integrated with the Autodesk Inventor software application. This interface is used to manage the complex relationships among Autodesk Inventor files and communicate this information to the vault.
project file	The locations and settings that identify all the information necessary for a set of Autodesk Inventor design files to resolve their relationships. The file extension is *.ipj*.
library	A collection of Autodesk Inventor design files that are commonly used in designs and infrequently modified.
Check In	Placing Autodesk Inventor files in the vault using the Inventor add-in. Always use the Inventor add-in to add Autodesk Inventor files to the vault to retain file and data relationships.
vault-enabled project file	An Autodesk Inventor project file configured for use with Autodesk Vault. The multi-user setting is set to Vault and the paths are modified as necessary.

Term	Definition
vault browser	A specialized browser added to the Inventor browser panel for working with files in a vault. The vault browser is available once Autodesk Vault is installed and the current project is vault enabled.
workspace	Within a vault project, the logical location for storing Autodesk Inventor files for a design. Set up the file structure of the workspace folder to be the same as the file structure of the local working folder.

Vault Add-in for Autodesk Inventor

The vault add-in works within Autodesk Inventor, and adds tools for managing files using the Autodesk Inventor interface. Through the add-in, you can add files to the vault, check in and check out files, and map folder locations. The add-in works only with Autodesk Inventor files. We recommend you use the add-in to place Autodesk Inventor design files in a vault so that relationships among the files are preserved.

When the Autodesk Vault client is installed, the vault browser is added to the Autodesk Inventor browser bar and the vault submenu is added to the Autodesk Inventor file menu.

Organizing Files

Typical designs might include parts, libraries, iPart Factories, purchased parts, and common manufactured parts. To organize your files for effective data management, it is helpful to understand how Autodesk Inventor finds these files using the workspace and library search paths in the Autodesk Inventor project file, and how the Autodesk Inventor Content Library and iPart Factories publish parts.

The workspace path in an Autodesk Inventor project file indicates the location of the design data. In a vault-enabled project file, the workspace path specifies the location of the design data in the vault relative to the project file.

Specifying Workspaces in Vault Projects

In the vault project file, define the workspace path as a period (.), meaning the same folder as the project file, or to a subfolder below the project file, such as .*Workspace*.

NOTE The best practice is to set the workspace path to the same folder as the project file.

For large design projects, the workspace folder can contain additional sub folders. However, you cannot map a project search path to a workspace sub folder.

For more information about setting up frequently used folders, see Autodesk Inventor Help.

Specifying Libraries in Vault Projects

Libraries, including Content Center libraries, contain files that are used in multiple designs. Library files do not change often and are frequently reused. For example, common components that are purchased, such as fasteners or electric motors, can be stored in a library.

Library files are typically stored in a library folder separate from the project files on a shared server so that all members of a design team have access to them. Library files cannot be modified in a vault-enabled project. Libraries cannot be located within the cone of influence.

Creating Folder Structures in Vaults

The master files for a vaulted project are all stored in the vault. To modify the files, they must be checked out to the local working folder. For ease of data management, the vault folder structure is set up to match the working folder structure for the project. For example, the vault folder structure for a project called Clamp is set up parallel to the structure of the local working directory and workspace folder.

Library folders cannot be located in the same folder structure as the model files in the vault. You add libraries to a vault at the root level of the vault directory. A project file must include a library search path for each of the libraries used by the model.

Working with Legacy Projects

Design projects that were created before Autodesk Vault was installed can either be converted to vault projects or you can use Autodesk Autoloader to create the vault project file. It is recommended you use Autodesk Autoloader to create the vault project. Autodesk Autoloader uploads an entire Autodesk Inventor project to a vault located on an Autodesk data management server.

To access Autodesk Autoloader: Program Files ➤ Autodesk ➤ Data Management Server ➤ Tools ➤ Autodesk Autoloader. Learn more about Autodesk Autoloader by clicking on Help.

NOTE If you choose to convert a legacy project to work with the vault using the Project Editor, back up the project file and data.

Converting Projects to Work With Vaults

Existing Autodesk Inventor projects can be converted for use with Autodesk Vault. To convert an existing project, you use the Project Editor to modify the project file to work with the vault and add the design data to the vault.

For more information about modifying a project file to be vault compatible, see "Converting Existing Project Files" in the Autodesk Vault Help.

Tips for converting projects to work with vaults

- In a shared or multi-user project, set the Multi-user setting to Vault.
- Consolidate Autodesk Inventor design files and the associated *.ipj* file manually.
- Before you remove a reference in the *.ipj* file, relocate the necessary design files from that location to the one referenced in the vault enabled *.ipj* file.
- Set up libraries in the vault for common components that are shared across multiple Vault Projects.
  Include libraries for iParts and Autodesk Inventor Content Libraries that are referenced by the project.

  In the Autodesk Inventor Projects dialog box, verify that the Autodesk Inventor Project File references the vault libraries correctly.
- For multi-user projects, make sure all users have their files checked in. All current data must be in the workgroup.

NOTE When converting an existing project to a vault-enabled project, turn on Using Unique File Names in the Autodesk Inventor project file. For more information, see Autodesk Inventor Help.

Moving Existing Data into Vaults

Existing Autodesk Inventor design data can be added to the vault. The data can be associated with a legacy Autodesk Inventor project file that has been converted to a vault project file or used with a brand new vault project file. In the Autodesk Vault add-in for Inventor, use the command Add Project to add existing data to the vault.

For more information about adding a project to the vault, see "Adding a Project" and "How to Add Files (Inventor)" in the Autodesk Vault Help.

NOTE It is recommended you use the utility called Autodesk Autoloader for uploading Inventor data into a vault instead of manually modifying the project file with the Project Editor. Autodesk Autoloader is a stand-alone utility independent of Autodesk Vault. Autoloader is available on the installation media. Contact the vault administrator for more information about installation and usage.

Logging into Vaults in Autodesk Inventor

If a vault is not open, the vault browser displays yellow icons with exclamation points in front of each file. When you log into the vault, the icons change to indicate the current status of the files, and the vault features are available on the context menu.

Log into a vault

1. Start Autodesk Inventor.
2. Click File ➤ Autodesk Data Management Server ➤ Log in.
3. **User name** The name for the vault account.
   Password The password associated with the vault account.
   Server The name of the computer where the vault server is installed.
   Database The name of a database located on the specified server. The default is Vault. Click the browse button

to select from a list of available databases on the server.

4 A default user ID, vault server, and database are set up for you to get started using the vault right away. If a user account has already been created for you, enter your account information. If a user account is not yet created for you or the default values are not present, contact the vault administrator.

5 In the Log in dialog box, enter your user name, password, the name of the vault server, and the name of the vault database.

6 Click OK.

Setting Up Vault-Enabled Projects

Autodesk Vault can manage only vault-enabled project files. When you start a new Autodesk Inventor model, create the vault-enabled project file before you create any model files. Configure the project file to take the maximum advantage of Autodesk Vault.

Set up library folders in the vault to manage reusable common components, iParts, and content libraries. You must map the library search paths in the project file to the library folders in the vault.

For more information about mapping, see Mapping Folders on page 44.

Autodesk Inventor project files include several settings that tell Autodesk Inventor where to look for model files. When Autodesk Inventor opens the components of an assembly, it looks in the folders specified in the project file. For example, if a component is a non library component, Autodesk Inventor looks in the workspace folder specified in the project file. If a component is a library part, Inventor looks in the library paths specified in the project file.

For more information about project files, see the sections about projects in the *Autodesk Inventor Getting Started* manual.

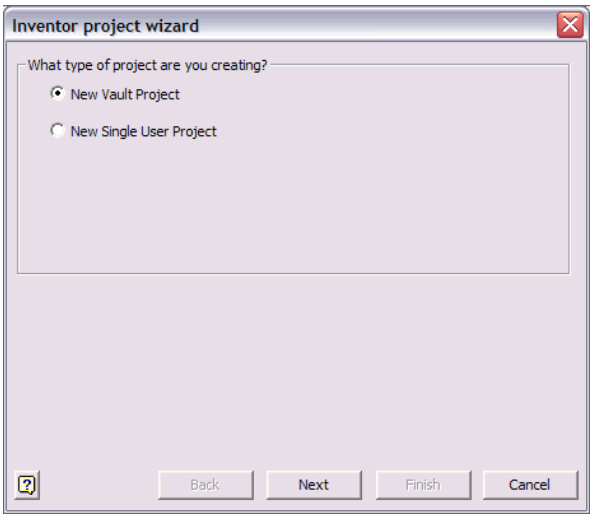
NOTE It is recommended you use Autodesk Autoloader to create the vault project. Autodesk Autoloader uploads an entire Autodesk Inventor project to a vault located on an Autodesk data management server.

Set up a new project file for a vault

1 Start Autodesk Inventor.

2 In the Open dialog box, click Projects ➤ New.

3 In the Autodesk Inventor project wizard select New Vault Project.



4 Click Next.

5 Name the vault project and specify the project location.

6 If you are referencing existing part libraries in the new project, click Next and select the libraries for the project.

If you are not referencing libraries, or you will be using new libraries, click Finish instead.

Specify a new library path

1 Click File ➤ Projects.

2 Double-click the new vault project to make it the active project.

3 Enter the library search paths where the library components are located on the network.

4 Click Save. Click Cancel to close the dialog box.

The new project file is now vault-enabled. As you create files in the project, you can add them to the vault using the Vault add-in for Autodesk Inventor.

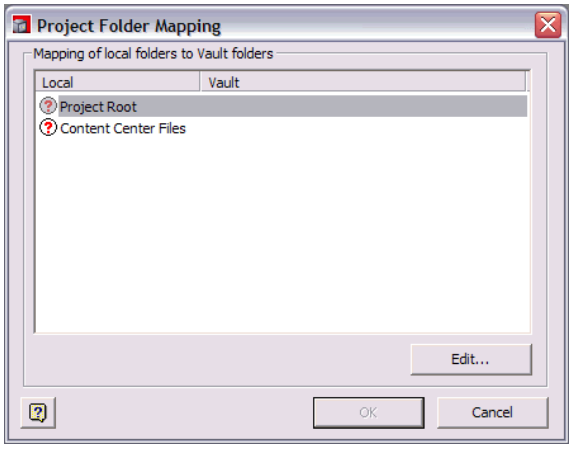
For more information about adding files to the vault from Inventor, see Working with Files in Vaults on page 46.

Mapping Folders

After the vault enabled project file is created, map the project root and any referenced libraries to corresponding folders in the vault.

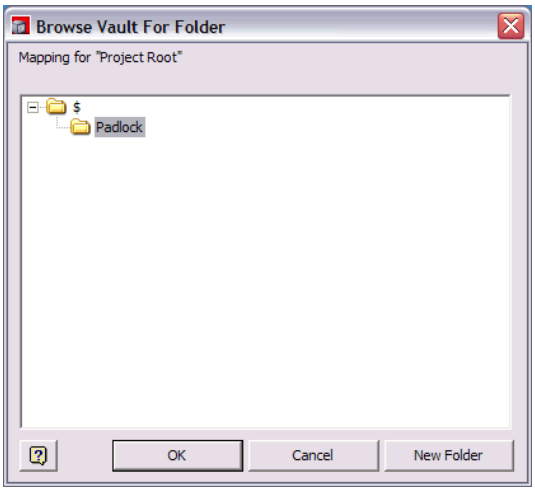
Map the project root

1. If you are not currently logged into the vault, click on the File menu in Autodesk Inventor ➤ Autodesk Data Management Server ➤ Log in.
2. Click File ➤ Vault ➤ Map Folders.
3. In the Project Folder Mapping dialog box, select Project Root, and then click Edit or double-click Project Root.



The Browse Vault for Folder dialog box displays a list of the folders in the vault.

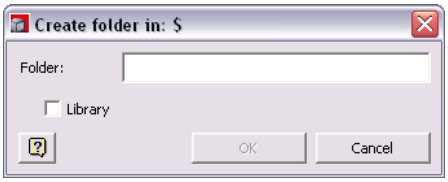
4. If the necessary folder exists in the vault, select the vault folder from the list and then click OK.



The existing vault folder is mapped to the project root.

5 To add a new folder to the vault, select the root of the vault, $, and then click New Folder.

In the Create Folder dialog box, enter a name for the new vault folder.



6 Click OK.

7 In the Browse Vault For Folder dialog box, select the new vault folder.

Click OK.

8 Click OK to exit the Project Folder Mapping dialog box.

The new vault folder is mapped to the project root.

When you map a vaulted library to a project, turn on the Library check box when you create the library folder. You can only create library folders directly off the root level of the vault.

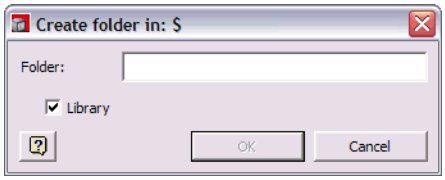
Map a library in a vault

1. If you are not currently logged into the vault, click the File menu in Autodesk Inventor ➤ Autodesk Data Management Server ➤ Log in.
2. Click File ➤ Vault ➤ Map Folders.
3. In the Project Folder Mapping dialog box, select the unmapped library. Click Edit.

   The Browse Vault for Folder dialog box lists the folders currently in the vault.
4. If the necessary folder exists in the vault, select the vault folder from the list and then click OK.

   The existing vault folder is mapped to the selected library.
5. To add a new folder to the vault, select the root of the vault, *$*, and then click New Folder.

   In the Create Folder dialog box, enter a name for the library folder, and select the Library check box.



6. Click OK.

   A library folder is added to the vault.
7. Select the new library folder. Click OK.
8. Click OK to exit the Project Folder Mapping dialog box.

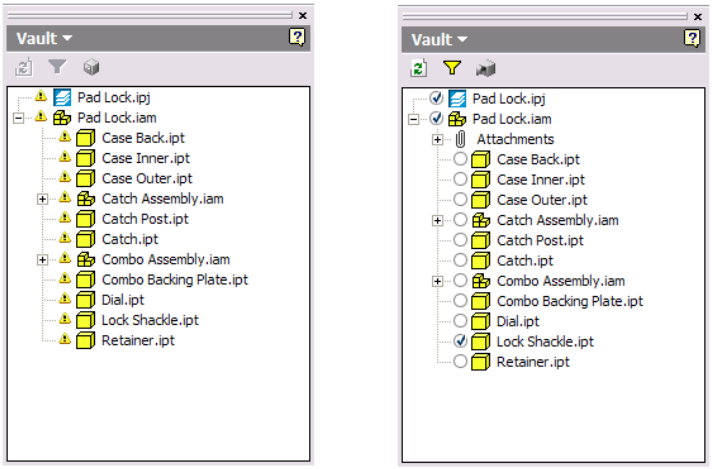
   The new vault library folder is mapped to the selected library.

Working with Files in Vaults

When you work on a file that was checked out of the vault, you are working on a local copy of the file and not the original file. At no point do you ever work on the actual vaulted file. When a modified file is checked into the vault, the modifications are available as the latest version in the vault. All past versions of a file are maintained in the vault.

Vault Browser

When Autodesk Vault is installed, the vault browser is added to Autodesk Inventor. The vault browser displays the status of model files in a vault and provides quick access to Autodesk Vault.



Filter the vault browser

- Click the filter and then select criterion from the list to show only files that are:
  - Show Non-Vaulted Only
  - Show Checked Out Only
  - Show Checked Out To Others Only
  - Show Modified Files Only

Log into a vault

- Click the vault status icon in the vault browser, and then select Log in.

Log out of a vault

- Click the vault status icon in the vault browser, and then select Log out.

Launch Vault

- Click the vault status icon in the vault browser, and then select Launch Vault.

Checking in Autodesk Inventor Files

You can store any type of design data in a vault, including documents, spreadsheets, and model files. Non-Autodesk Inventor files are added to the vault using Autodesk Vault. To maintain the relationships among Autodesk Inventor model files, you must use the Vault add-in for Autodesk Inventor to add them to the vault.

A question mark icon is displayed next to Autodesk Inventor files that are not currently in the vault.

Add/Check in files to a vault from Autodesk Inventor

1. On the Autodesk Inventor browser bar, select vault browser.
2. Right-click a file in the vault browser, and then click Check In.
3. In the Add Files dialog box, the list of files to be added to the vault is displayed.

   You can add comments describing the files.
4. If you want to add the files to the vault and keep them checked out so you can continue working on them, select Keep Checked Out.
5. Click OK.

When Autodesk Inventor files are added to the vault, the file relationships are recorded as well. The status of each file added to the vault is indicated by the icon next to the file name in the vault browser.

NOTE Any sub-folders required below the project file are created automatically.

Adding Projects

You can manually add an entire Autodesk Inventor project to a vault in a single operation. All drawings, parts, assemblies, and presentations for the active project, as well as any support files within the project structure such as documents and spreadsheets, are added to the vault at the same time.

NOTE Autodesk Inventor project files that are not vault-enabled cannot be added to the vault. The project file must be vault-enabled and the project folders must be mapped to vault folders prior to adding the project.

Library files that have not been added to the vault are added if the library path in the project file is mapped to a vault folder. Library files are the only files located outside of the project structure that are added to the vault using the Add Project command.

Each time you use the Add Project command for the active project, any new files contained within the local project structure that are not yet in the vault are added to the vault.

The Add Project command maintains the relationships among the files being added, including:

- Assembly files (*.iam*) including relationships with *.idv* files
- Part files (*.ipt*)
- Drawing files (*.idw*)
- Presentation files (*.ipn*)

Add a project to a vault

1. Click File ➤ Vault ➤ Add Project. If the vault folders are mapped to the project, the local files are scanned for relationships.
2. In the Add Project dialog box, verify that the project structure is complete in the file list.
3. Click Settings to specify whether or not *.dwf* files are created when files are added to the vault. The *.dwf* attachments are only created for files that have changed or for files that do not have *.dwf* files published already.
4. Enter a comment for the files.
5. Click OK.

The files are added to the vault.

NOTE Use Add Projects to add IDW files or an entire project.

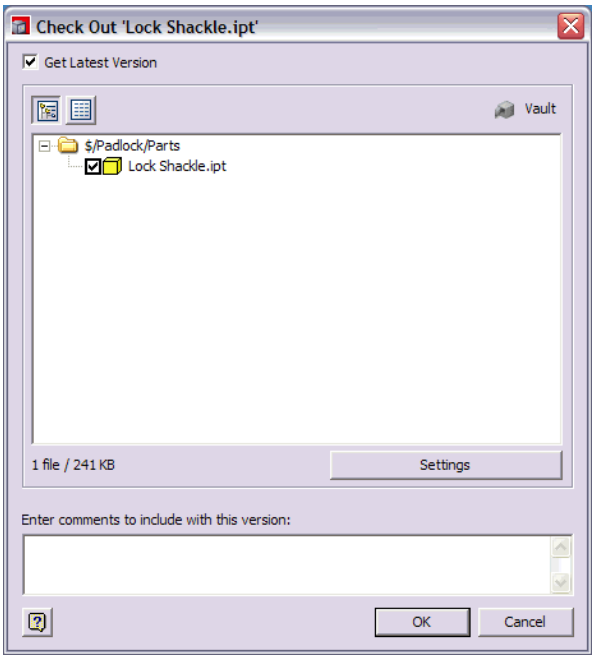
NOTE It is recommended you use Autodesk Autoloader to create the vault project. Autodesk Autoloader uploads an entire Autodesk Inventor project to a vault located on an Autodesk data management server.

Checking Out Files

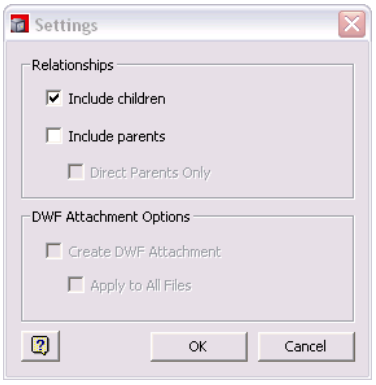
To make changes to a file, you must first check it out of the vault. No one else can check out a file that is checked out until it is checked back in to the vault. You can check out one or any number of files.

Check out a file from Autodesk Inventor

1. In the vault browser in Autodesk Inventor, right-click, and then click Check Out.
2. In the Check Out dialog box, enter a comment and specify whether or not to get the latest version.



Click Settings to specify whether or not *.dwf* files are created when files are added to the vault. The *.dwf* attachments are only created for files that have changed or for files that do not have *.dwf* files published already.



3 Click OK.

A read/write copy of the selected file and any specified dependencies are placed in the local working folder. The status of the checked out files is indicated in the vault browser.

Checking In Files

As you work on Autodesk Inventor files, changes are saved to the local copy. To record the changes in the vault, the file must be check the file in.

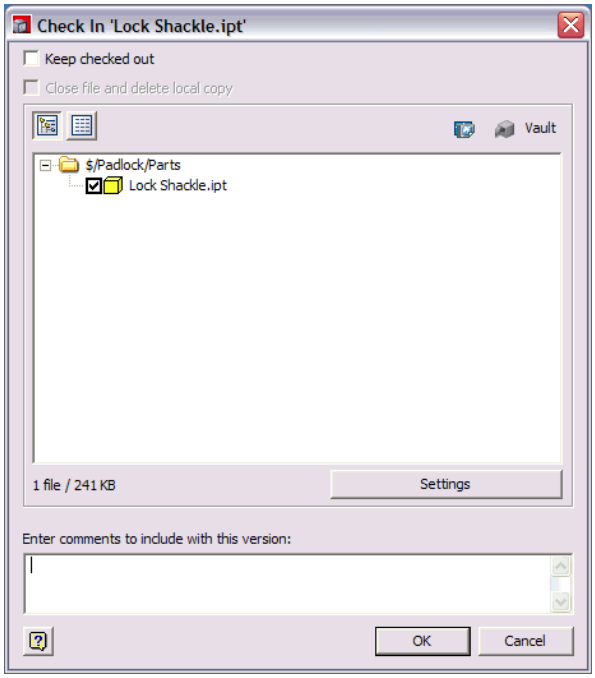
Check in a file from Autodesk Inventor

1. Save the file.

   If the file is not saved, you are prompted to save the changes.

2. In the vault browser in Autodesk Inventor, right-click an Autodesk Inventor file, and click Check In.

3. In the Check In dialog box, enter a comment and specify whether or not to keep the file checked out for further editing.



If you are checking in an entire assembly, a drawing, or a presentation, you can choose to close the file and delete the local copy. If you choose not to check in all the components of the assembly, this option is not available.

Click Settings to choose whether or not parents and children of the file are also checked in at the same time and to specify whether or not *.dwf* files are automatically created when files are checked in. The *.dwf* attachments are only created for files that have changed or for files that do not have *.dwf* files published already.



4 Click OK.

The files and specified relationships are checked back into the vault. The status of the checked in files indicates that the files are in the vault.

Undoing File Check Outs

You can undo a file check out. Undo Check Out removes the check out status of the file in the vault and the file is restored to the latest version that was checked in. You can choose whether or not the local copy is replaced with the version in the vault.

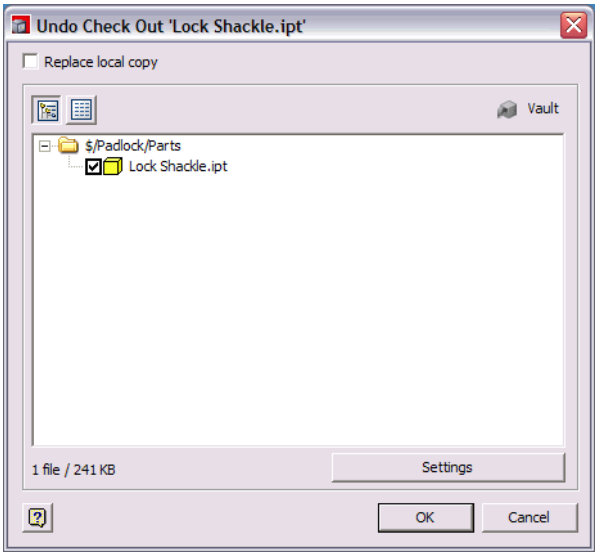
Undo a check out in Autodesk Inventor

1 In the vault browser in Autodesk Inventor, right-click an Autodesk Inventor file, and then click Undo Check Out.

2 In the Undo Check Out dialog box, specify whether or not to replace the local copy with the latest version in the vault.

If you replace the local copy, changes made to the local copy are lost.

Click Settings to choose whether or not parents and children of the file are returned to the vault without saving any changes.



3 Click OK.

Getting Latest Versions

You can retrieve the latest versions of parts and assemblies from the vault that are not checked out to you. Use this method to synchronize the models in your current Autodesk Inventor session with the rest of the design team and to keep your assembly current. Files that are checked out to you and local files that have changed are not updated. Files modified by other users are updated to the latest versions in the vault.

Get the latest version of a model in Autodesk Inventor

1 In Autodesk Inventor, in the vault browser, right-click, and then click Get Latest Version.

You are prompted to verify that the files in memory are going to be updated.

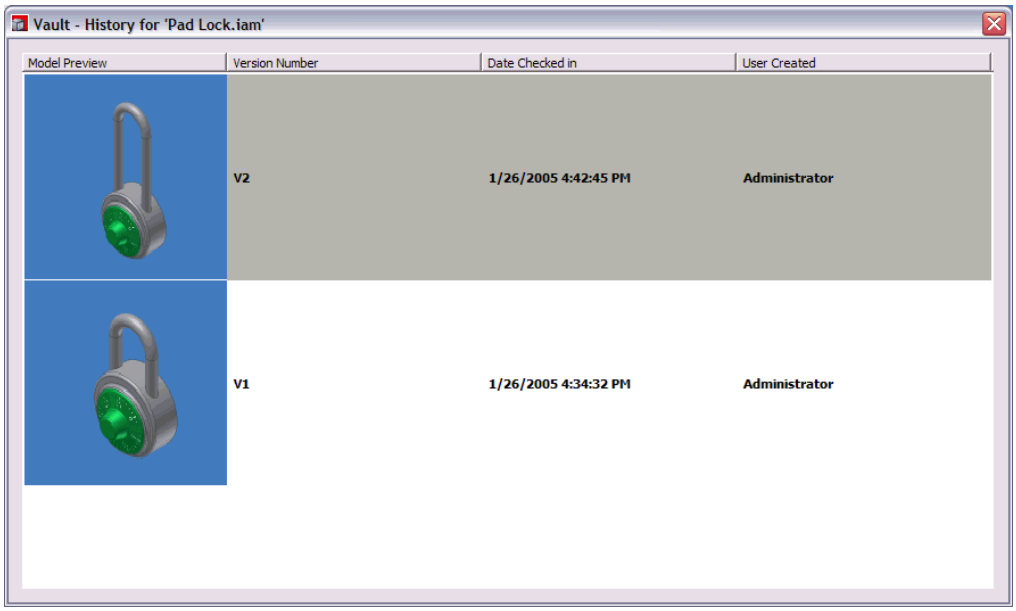
2 Click OK.

Viewing Histories of Files

The history shows the versions of files that exist in the vault, the date each version was checked in, and the authors of the modifications.

View the history of a file in Autodesk Inventor

- In the vault browser, right-click a file, and then select Show History. The history of the file is displayed in a dialog box.



Starting Autodesk Vault

When you start Autodesk Vault from inside Autodesk Inventor, you have access to the vault features and the features in the Vault add-in for Autodesk Inventor.

Start Autodesk Vault from Autodesk Inventor

1. On the File menu in Autodesk Inventor, click Vault ➤ Launch Vault.
   The Log in dialog box is displayed.

2 Log into the vault.

Autodesk Vault displays the contents of the opened vault.

NOTE You can also click the Vault button in the vault browser to launch Vault.

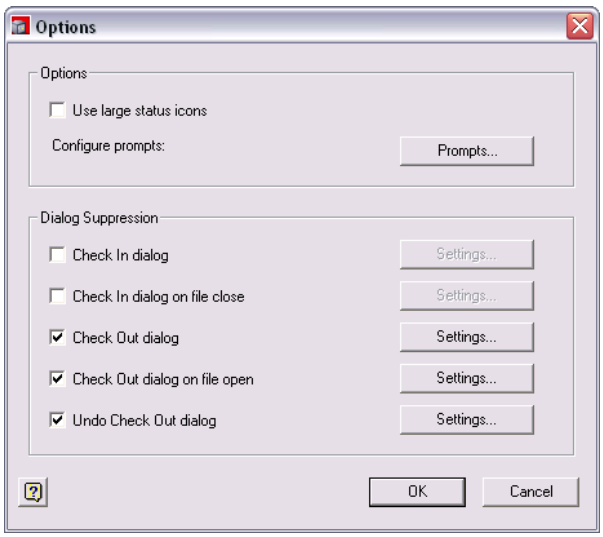
Customizing Work Flows

You can customize the work environment for vault operations in Inventor. You can greatly streamline your workflow by suppressing dialog boxes for certain commands and controlling the default responses to various prompts, as well as when and how often prompts appear. For more information on customizing the vault workflow within the Inventor environment, see the Autodesk Vault for Inventor Help.

Customize your work environment

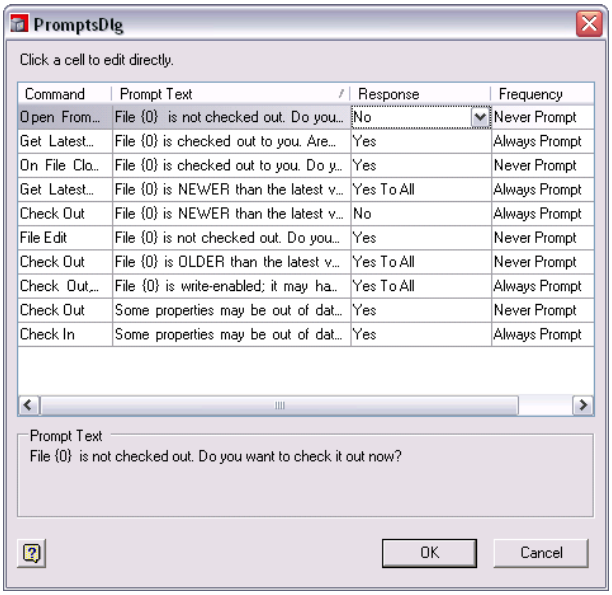
1 Select File ➤ Vault ➤ Options.

2 In the Options panel you can choose whether or not to display large or small status icons by clicking the Large status icons check box.



3 To configure default prompts, click Prompts. In the displayed list, each row represents a prompt. To modify default responses to prompts, click

on the drop down menu in the Response cell. Use the Frequency cell to adjust when and how often a prompt appears.



4 In the Dialog Suppression panel, there are five dialog suppression options available. To suppress a dialog box click the check box next to it. The Settings button will become available. Click the Settings button and adjust the default settings.

5 Click OK.

Using Vault with AutoCAD Products

5

Learn how to use Autodesk® Vault with AutoCAD®, AutoCAD Mechanical®, Autodesk® Mechanical Desktop®, and AutoCAD® Electrical.

Key Terms

Term	Definition
AutoCAD Vault Add-in	The Autodesk Vault client that is integrated with AutoCAD, Mechanical Desktop, AutoCAD Mechanical, and AutoCAD Electrical software. This interface is used to manage the complex relationships among the files and communicate this information to the vault.
Project Manager	Vault operations are performed in an AutoCAD Electrical project using the Project Manager.
Xref Manager ESW	The Xref Manager Enhanced Secondary Window serves as the interface by which all vault operations can be performed and managed. Standard Xref operations are also available.
check in for the first time	Checking in a DWG or image file to the vault for the first time. Adds a file to the vault so the file can be managed by the vault.
check in a file back into the vault	Returning a modified file to the vault. The previous version is retained in the file history.

Term	Definition
check out	Retrieving a read/write copy of a file stored in the vault and downloading the file to the working folder on the local drive for editing.
refresh from the vault	Compares the file in the vault to the corresponding working copy on your local drive and updates the vault status icons (AutoCAD, Mechanical Desktop and AutoCAD Mechanical only).
reload from the vault	Downloads one or more files from the vault to your working folder on your local drive and reloads them into AutoCAD.
latest version	The most recent version of a file stored in the vault.
local copy	The copy of the vault source file placed in the local working directory.
previous version	An older version of a file stored in the vault. All previous versions of a file are retained and accessible at any time from Autodesk® Vault.
working folder	A local folder to which files are copied from the vault for editing. A working folder must be mapped to the root *$*.
($)	The root of the vault. Analogous to *C:* on the local computer.

About the Vault Add-in for AutoCAD

The AutoCAD®Vault add-in works within AutoCAD, AutoCAD Mechanical, Mechanical Desktop, and AutoCAD Electrical, adding data management tools to the interface. Through the AutoCAD Vault add-in, you can add files to a vault, and check files out and in. The add-in works with both DWG and image files. The recommended method for performing vault operations depends upon your working environment.

For more information about performing vault operations with AutoCAD Electrical files, see AutoCAD Electrical on page 77.

When the add-in is installed, the Xref Manager becomes an Enhanced Standard Window (ESW) that supports both drawing xrefs and image files in the same window. The Xref Manager supports all of the vault operations and status

icons. The vault commands are only available when you are logged into the vault. You can also access the vault commands from the file menu and the command line when you are logged into the vault.

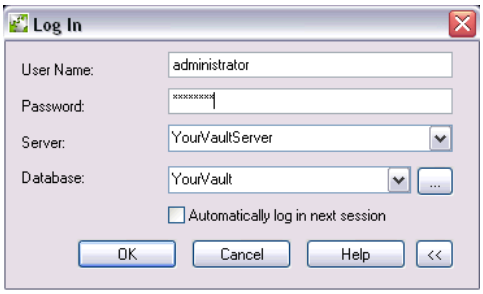
NOTE To support vault commands in SDI (Single Document) mode, set the SDI system variable to "0."

Performing Tasks In Vaults

To access a vault from inside AutoCAD, AutoCAD Mechanical, Mechanical Desktop, and AutoCAD Electrical, you must log into the vault. Use the vault icon located in the AutoCAD tray to log in. When you are not logged into the vault, the icon appears as a closed safe. After you log in, the vault icon changes to an open safe. The tool tip on the vault icon indicates the login status and the server\database you are currently logged into.

Logging into Vaults in AutoCAD-based Products

1. Start AutoCAD, AutoCAD Mechanical, Mechanical Desktop, or AutoCAD Electrical.
2. Use one of these methods to open the Login dialog box:
   - Click File ➤ Vault ➤ Log in.
   - In the AutoCAD tray, right-click the vault icon, and then click Log in.
   - At the command line, enter vault. Press ENTER. Enter *login*. Press ENTER.
   - On the Vault toolbar, click the Log In tool.
   - In the Xref Manager window, right-click the background, and click Log in.
3. In the Log In dialog box, log in.



4 Click OK.

NOTE The names of the vault server and vault database are stored in your system registry. The user name and password can be saved for quick login. Click Automatically login next session to automatically log into the vault using the current account the next time you start Autodesk Vault.

The vault server stores a unique copy of each version of a file in its file store. These files stored in the vault are called the master files or master copies.

Setting Working Folders on Local Drives

The working folder in the vault specifies the folder to which files are added and from which they are checked out. The vault server copies files from folders defined in the vault to folders specified as their working folders. You must set a working folder to perform any action that copies a file out of the vault.

NOTE For more information about mapping a working folder, see Integrating with Autodesk Inventor.

---

1 Start Autodesk Vault in AutoCAD, AutoCAD Mechanical, Mechanical Desktop, or AutoCAD Electrical, using one of these methods:

- On the File menu, click File ➤ Vault ➤ Launch Vault.
- Double-click the vault icon in the AutoCAD tray.

2 In Autodesk Vault, click File ➤ Set Working Folder. The working folder can be set only at the root of the vault (*$*).

3 In the Browse for Folder dialog box, browse to the working folder on your local computer where you plan to store the files you check out of the vault.

4 **NOTE** The Working Folder is set only once. Plan the folder structure before you start.

5 If you do not set a working folder for *$*, then the first time you add a file to the vault in AutoCAD, AutoCAD Mechanical, Mechanical Desktop, or AutoCAD Electrical, a working folder named Vault is set on the local drive (*My Documents/Vault*).

NOTE You may not need to set a local working folder if your system administrator pre-set a consistent working folder on the network for all the users on the same vault to use.

Creating Folders for Vault Projects

Use folders to organize files in the vault in the same way you organize files on the your computer. You can use Autodesk Vault to create the folder structure for the vault or you can do so within AutoCAD, AutoCAD Mechanical, Mechanical Desktop, or AutoCAD Electrical. The root directory in a vault is represented as *$/*.

NOTE You can define library folders for read-only parts. For more information about creating library folders and working with library files, see "Working with Library Files".

1 Right-click *$*, and then click New Folder.

2 Enter a descriptive name for the vault project. Click OK.

You can create a folder off of the root (*$*) for each additional project. The root working folder remains the same.

Working with Files in Vaults

When you work on a file that is checked out of the vault. you work on a local copy of the file and not the original. At no point do you ever work on the actual file stored in the vault. When a modified file is checked into the vault,

the modifications are available as the latest version in the vault. All past versions of a file are maintained in the vault.

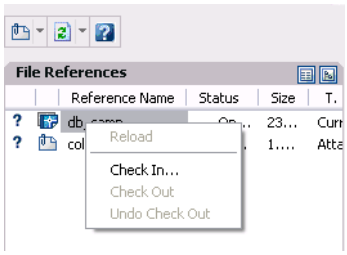
Adding Files

Use the Check In command to store DWG and image files in a vault. Use Autodesk Vault to add non DWG files to the vault. It is recommended that you add AutoCAD, AutoCAD Mechanical, Mechanical Desktop, and AutoCAD Electrical files to the vault from within the respective program.

Use the Check In Folder command to check in an entire folder or to add files to the Vault for the first time. The Check In Folder command is used to check in AutoCAD Electrical projects.

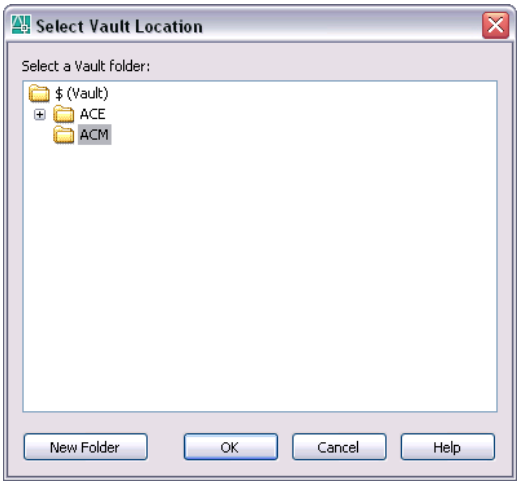
Check in a file for the first time

1. In AutoCAD, AutoCAD Mechanical, or Mechanical Desktop, open the file to add to the vault. For more information about performing vault operations with AutoCAD Electrical files, see AutoCAD Electrical on page 77.
2. Use one of these methods:
   - Click File ➤ Vault ➤ Check In.
   - At the command line enter *vault*. Click ENTER. Enter *check in*. Click ENTER.
   - On the Vault toolbar, click the Check In tool.
   - In the Xref Manager, right-click the file name, and then click Check In.



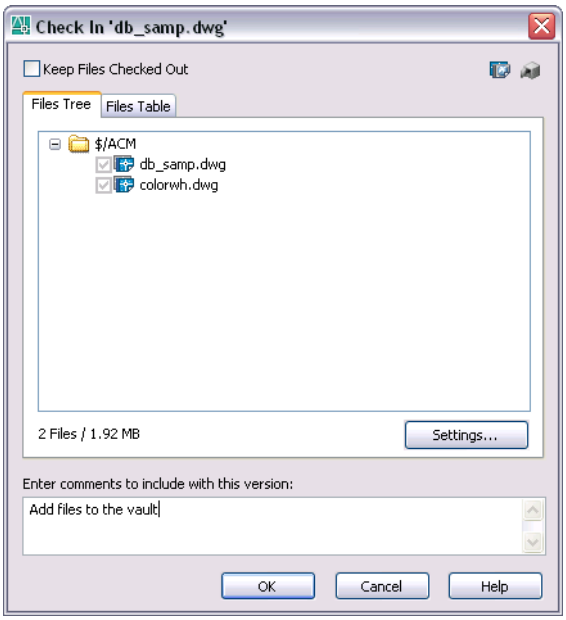
NOTE If the active DWG is a host file containing xrefs, the xrefs are listed in the Check In dialog box.

3 In the Select Vault Location dialog box, specify where to add the new files.



4 To create a new folder in the Select Vault Locations dialog box, click New Folder. Create a new folder at the root ($) or on an existing folder.

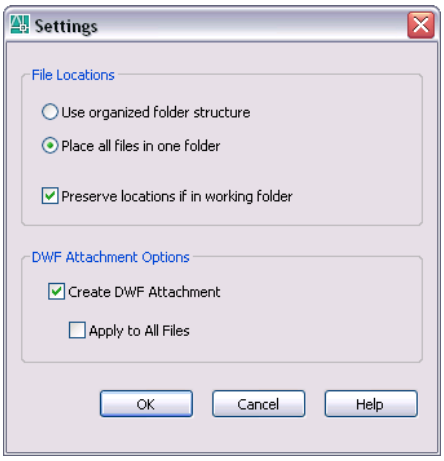
In the Check In dialog box, all files that are referenced into the active file are displayed.



5 Click Settings ➤ Place all files in one folder.

6 The File Locations settings determine how files are added to the vault the for first time.

The Create DWF Attachment settings specify whether or not *.dwf* files are created when files are added to the vault. The *.dwf* attachments are only created for files that have changed or for files that do not have *.dwf* files published already.



7 Click OK.

8 Click OK to exit the Check In dialog box.

9 When you add a host file and associated xrefs to the vault, the relationships are recorded and maintained. The status of each file added to the vault is indicated by the icon next to the file name in the Xref Manager.

10 Any subfolders required below the host file are created automatically.

Checking Out Files

To make changes to a file, you must first check it out of the vault. A file that is checked out of the vault cannot be checked out by anyone else until it is checked back in. In a host file containing xrefs, you can check out one file, multiple files, or all referenced files within the Xref Manager. Files that do not contain xrefs can only be checked out one at a time.

1 In AutoCAD, AutoCAD Mechanical, Mechanical Desktop, or AutoCAD Electrical, use one of these methods to check out a file:

- Click File ➤ Open from Vault.
- On the Vault toolbar, click the Open from Vault tool.
- On the Vault toolbar, click the Check Out tool.

- In the Xref Manager, right-click one or more xrefs, and select Check Out (only available in a file stored in the vault that contain xrefs).
- In AutoCAD Electrical, right-click the file in the Project Manager, select Check Out.

2. If you are not logged into the vault you are prompted to log in.
3. In the Select File dialog box, select a vault folder and highlight a file name.
4. Click Ok, or click the arrow on the Open button and select one of the following options:

Open (Check Out)	Checks out the drawing file that opened (default).
Open (Check Out All)	Checks out the drawing files that open and all dependents (only available in host files that contain xrefs).
Open (Read Only)	Opens the file in a read only state without checking it out of the vault.

Click OK.

A read/write copy of the selected file and any specified dependencies that are checked out are placed in the local working folder. The status of a checked out file is indicated in the Xref Manager.

Working with Library Files

Library files do not change often and are frequently reused. When you check a file into a library folder in the vault, it becomes a library file. You can create new library folders in Autodesk Vault or in the Select Vault Location dialog box in AutoCAD, AutoCAD Mechanical, or Mechanical Desktop.

Create a library folder

- In Autodesk Vault, right-click the root level directory, and then click New Library Folder.

NOTE You can create a new Library folder in the Select Vault Location dialog box in AutoCAD, AutoCAD Mechanical, or Mechanical Desktop when you check a host file into the vault for the first time.

Add a library file in Autodesk Vault

1. In Autodesk Vault, right click the root (*$*) and then click New Library Folder.
2. Enter a descriptive name for the new library folder.
3. In Windows Explorer, browse to the file you want to add, and drag it to the new library folder.
4. In the Add files dialog box, the file you want to add is displayed.
5. Click OK.

Attach a library file from the Vault to a drawing

1. Start with a file from the vault opened in AutoCAD, AutoCAD Mechanical, or Mechanical Desktop.
2. Use one of these methods to attach a file from a vault.
   - In the Xref Manager, right-click the background, and then click Attach from Vault.
   - On the toolbar in the Xref Manager window, click Attach, and then select Attach from Vault.
   - On the Vault toolbar, click Attach from Vault.
3. In the Select file dialog box, double-click the Library folder. Highlight the library file name. Click Open.
4. Insert an instance of the library file into the current drawing.
5. Save the drawing, and then check the library file into the vault.

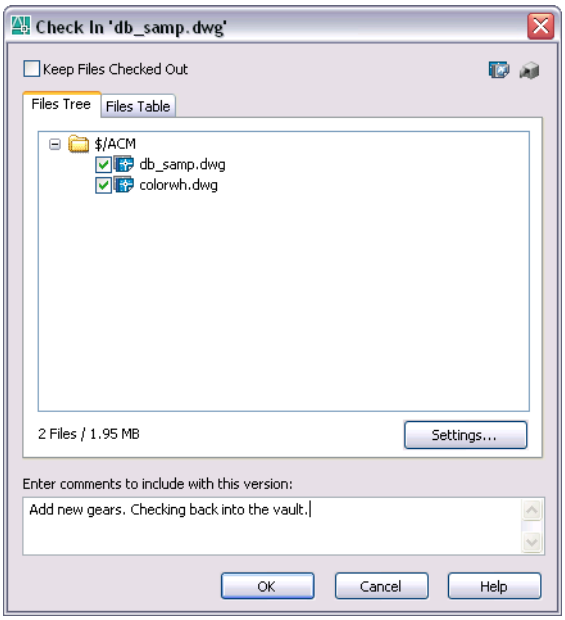
NOTE You must check the host file into the vault to create a relationship between the host file and the library file.

As you work on AutoCAD, AutoCAD Mechanical, Mechanical Desktop, or AutoCAD Electrical files, you can save changes to your local copy. However, to record the changes in the vault, you must check the file back in. You can check files back into the vault from within AutoCAD, AutoCAD Mechanical, Mechanical Desktop and AutoCAD Electrical.

Checking In Files

1. In AutoCAD, AutoCAD Mechanical, Mechanical Desktop or AutoCAD Electrical, save the files.
2. Check the files back into the vault using one of these methods:
   - On the File menu, click File ➤ Vault ➤ Check In.
   - On the Vault toolbar, click Check In.
   - At the command line, enter *vault*. Press ENTER on your keyboard. Enter *check in*. Press ENTER.
   - In the Xref Manager, right-click one or more files, and then click Check In.
   - In AutoCAD Electrical, right-click the file in the Project Manager, and then click Check In.
3. In the Check In dialog box, select Keep Files Checked Out to keep the file checked out for further editing.

   This checks in your changes to the vault so that others can update their files, and it keeps the file checked out to you. You can also enter a comment regarding the changes made to the file.



4 Click OK.

The files and specified relationships are checked back into the vault. You can view the status of files in the Xref Manager.

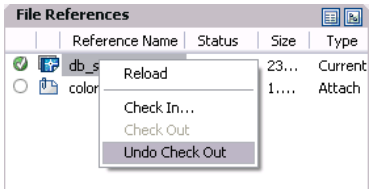
Undoing Check Outs

A file that is checked out can be returned to the vault without checking in changes. Undo Check Out releases your reservation on the file.

Undo a checkout

1 In AutoCAD, AutoCAD Mechanical, Mechanical Desktop or AutoCAD Electrical, undo the check out of one or more files using one of these methods:

- Click File ➤ Vault ➤ Undo Check Out.
- In the Xref Manager, right-click one or more files, and then click Undo Checkout.



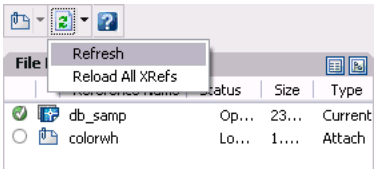
- On the Vault toolbar, click Undo Check Out.
- In AutoCAD Electrical, right-click the file in the Project Manager, and then click Undo Check Out.

Refreshing Models from Vaults

The Refresh from Vault command updates the status icons in the Xref Manager. Use this command to display the most current status information about the files in the vault project you are working on.

Refresh a model

1. In AutoCAD, AutoCAD Mechanical, or Mechanical Desktop, on the Insert menu, click Insert ➤ Xref Manager.
2. In the Xref Manager tool bar, click the Refresh tool to refresh the model.



3. Click OK.

Reloading Xrefs from Vaults

In files that contain xrefs, use the Reload all Xrefs command to download the latest version of the referenced files to the working folder on your local drive. Use this command to synchronize the models in your current AutoCAD, AutoCAD Mechanical, or Mechanical Desktop session with the rest of the design team and to keep your file up-to-date.

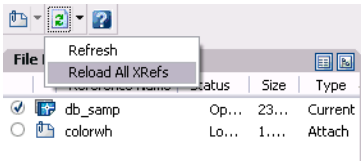
Files that are checked out to you and local files that have changed are not updated. Files modified by other users are updated to the latest versions in the vault.

You can cancel the download by clicking the Cancel button on the progress dialog box.

Reload an xref from a vault

1. Use one of these methods:
   - In AutoCAD, AutoCAD Mechanical, or Mechanical Desktop, in the Xref Manager list view, right-click one or more files, and then click Reload.
     To select multiple files, hold down SHIFT or CTRL as you select the files.

     To remove files from the selection set, hold down SHIFT or CTRL as you select the files.
   - To reload all xrefs from the vault, in the Xref Manager, Xref Manager toolbar, click the Reload All Xrefs tool.

   - To access the reload command, at the command line enter *-xref*. Press ENTER. Enter *reload*. Press ENTER. Enter the xref name(s) to reload.
2. At the prompt, verify that the files in memory are going to be updated.

Starting Vault

In AutoCAD, AutoCAD Mechanical, Mechanical Desktop, or AutoCAD Electrical, use one of these methods to start Autodesk Vault:

- Click File ➤ Vault ➤ Launch Vault.
- In the AutoCAD tray, double-click the vault icon.
- In the AutoCAD tray, right-click the vault icon, and then click Launch Vault.

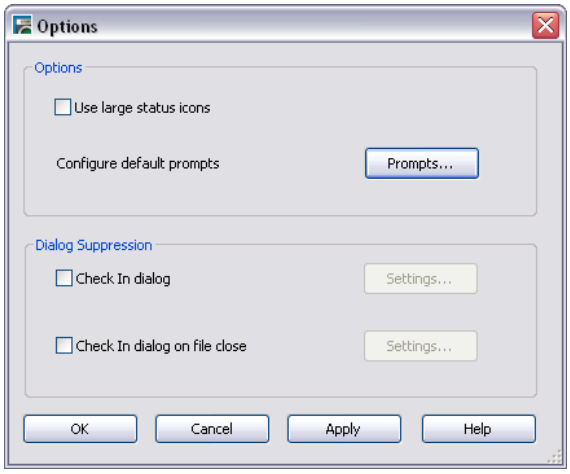
Customizing Work Flows

You can customize the work environment for vault operations with AutoCAD products, including AutoCAD, AutoCAD Mechanical, AutoCAD Mechanical Desktop, AutoCAD Electrical. You can greatly streamline your workflow by suppressing dialog boxes for certain commands and controlling the default responses to various prompts, as well as when and how often prompts appear.

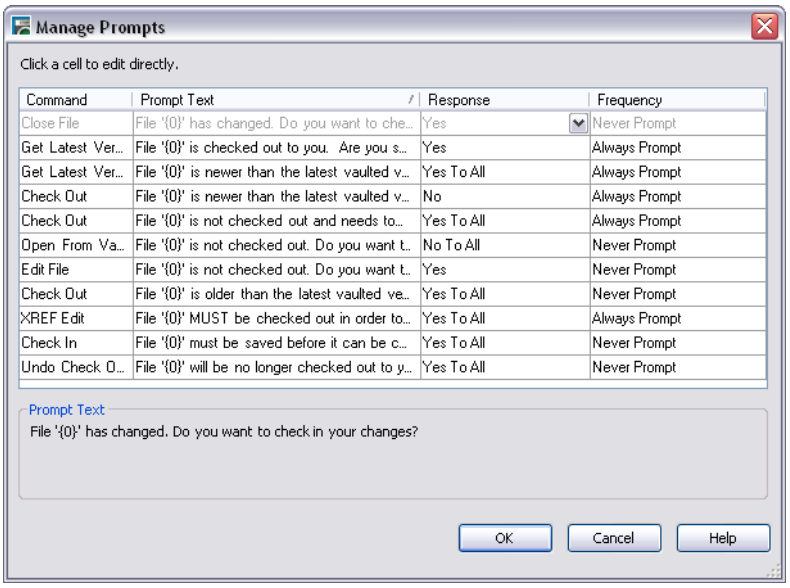
For more information on customizing your vault workflow within the AutoCAD add-in, see AutoCAD Vault Help.

Customize your work flow

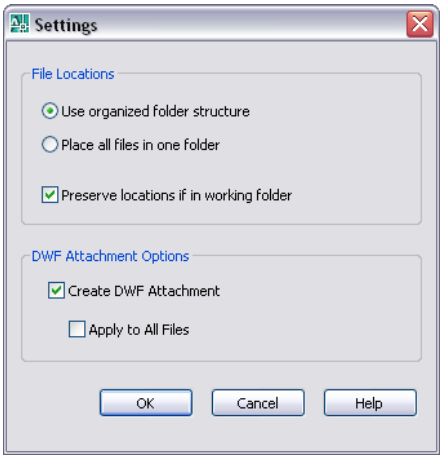
1 Select File ➤ Vault ➤ Options. In the options panel you can choose to display large or small status icons.



2 To configure default prompts, click the Prompts... button. In the Manage Prompts dialog box, each row represents a specific prompt. To modify default responses to prompts, click on the drop down list in the Response cell to select the desired response. You can also adjust how often each prompt should appear.



3 In the Dialog Suppression panel, you can choose whether or not the Check In dialog and Check In dialog on file close are suppressed. The Settings button will become available if you choose to suppress it. Click the Settings button and adjust the default settings. These settings will be used when the check in command is invoked.



4 Click OK.

Recommended Workflows

The workflows for AutoCAD Mechanical and Mechanical Desktop differ from those for AutoCAD Electrical.

AutoCAD Mechanical and Mechanical Desktop

Before you can add files to the vault you must set a working folder in Autodesk Vault. Launch Autodesk Vault within AutoCAD Mechanical to log into the vault and set a working folder.

In files that contain structured components, use the Externalize command to create xrefs of components you want to exist in separate files, such as:

- Components you want to track and control file versions.
- Components that are frequently reused.

You can perform all vault operations on the File menu, the Vault toolbar, at the command line, and within the Xref Manager. File status information is available in the Xref Manager.

Workflow overview

1. Start AutoCAD Mechanical or Mechanical Desktop.
2. Log into a vault.
3. If you did not set a working folder on the local drive, start Vault and set a working folder. Switch back to AutoCAD Mechanical or Mechanical Desktop.
4. Open a file you want to add to the vault.
5. Add the opened file to the vault using the Check In command. If the file contains xrefs, they are added to the vault when you check the host file into the vault for the first time.
6. Use the Open from Vault command to open and check out DWG and image files in the vault. If the file contains xrefs, you can check out one or more of the referenced files within the Xref Manager.
7. If you are working on a file containing xrefs, view the status of the files in Xref Manager. Use the Refresh and Reload command to get the latest versions of files that are out of date.

8 When you finish modifying the files, check them back into the vault. This is especially important when you are working on files that have a relationship to other files in the vault.

AutoCAD Electrical

In AutoCAD Electrical, you work on one project at a time. The project file (WDP) lists all the DWGs that are part of a project. When you make a change in one drawing, all files related to that drawing are automatically updated.

You can perform vault operations on the entire project or individual drawing files listed within the project in AutoCAD Electrical. However, in a multiple user design environment, you can choose to check out and edit individual files as they are needed rather than checking out the entire project at a time while still maintaining drawing file dependencies and versions. After you make changes to the files and check them back into the vault, the other associated files are simultaneously updated.

Use the Project Manager to perform all vault operations. In AutoCAD Electrical, you can select a file (or multiple files) within a project to:

- Check in (all)
- Check out (all)
- Undo check out (all)
- Get latest (all))

Workflow overview

1 Start AutoCAD Electrical.

2 Log into a vault.

3 In a single user environment, if you did not set a working folder yet, start Autodesk Vault and set a working folder on your local computer and then switch back to AutoCAD Electrical.

In a multiple-user design environment, set the working folder on a shared network resource for the entire project team.

4 Open a project you want to add to the vault.

5 Add the opened project to the vault using the Check In or Check In Folder command.

6 Use the Open from Vault or Check Out command to open and check out the file from the vault.

7 To work on more files in the project, check out multiple files to the working folder using the Check Out All command in the Project Manager.

8 When you finish modifying the files, check them back into the vault using the Check In All command in the Project Manager. All the related files will be updated.

Using Shared Workspaces

A shared workspace is a working folder located on a shared server for all users to access. The shared workspace configuration can be used with:

- Autodesk Vault
- Autodesk Productstream
- AutoCAD Vault Add-in
- Microsoft® Vault Add-in.

NOTE The Vault for Inventor Add-in does not support shared working folders.

You can choose to work in a local or shared working folder according to your design requirements. A shared working folder is highly recommended for the AutoCAD Electrical environment, especially in a multi-user situation, because it enables you and your design team to keep all files up-to-date.

For more information on configuring a shared workspace, see AutoCAD Vault Help.

Setting Shared Working Folders

To use a shared workspace, the system administrator should pre-set a consistent working folder for all project members to use. Assign the working folder location to the root level (*$*) of the vault. A shared working folder cannot be assigned to a subfolder.

NOTE If Inventor add-in clients will be accessing the same vault, do not enforce a shared working folder. If Inventor and AutoCAD Electrical are sharing the same vault, the vault administrator cannon enforce the shared working folder. Each AutoCAD Electrical user must set the working folder individually to point to a common network drive.

Rules For Using Shared Working Folders

Using a shared workspace means multiple users may be working on the same files. All your vault operations are protected as long as you log into the vault prior to working on the files. Always make sure you log into the vault before you do any operations to the files in a shared working folder.

The following guidelines will help prevent you from overwriting the changes made by someone else, and vice versa.

1. If a file is currently checked out to another user, you cannot perform the following operations:
   - Get Latest Version
   - Get Previous Version
   - Check Out
   - Undo Check Out
   - Reload (AutoCAD add-in only)

   This will prevent you from overwriting others' changes to that file.
2. You cannot check out a file that is currently opened for read-write by another user.
3. You can still check out a file that is opened for read-only by another user.
4. You can open a file in read-only when it is currently checked out to someone else using the same working folder.

For more information on working in a shared working folder, see AutoCAD Vault Help.

Working with Autodesk Productstream

6

Learn how to take data management to the next level using Autodesk® Productstream® to track the design release and manufacturing process through items, bills of material (BOM), engineering change orders (ECO), and the web client.

Key Terms

Term	Definition
approver	A change order participant who can approve or reject a change order.
assign item	Creates an item record in the item master associated with a CAD file stored in the vault.
change administrator	The creator of a change order. The change administrator cannot approve a change order.
effectivity	The date on which a change order either becomes effective or expires.
Engineering Change Order (ECO)	A documented request to implement a change to an item.

Term	Definition
item	A unique identifier that tracks all related data to a physical part, product, or assembly.
item master	The complete collection of all items in Productstream.
item numbering scheme	A pattern for numbering items.
lifecycle	Indicates where the item is within the design and manufacturing process.
responsible engineer	A change order participant who submits the change order for final approval
reviewer	A change order participant who can only read and comment on a change order. A reviewer can not approve or reject a change order.
revision	Indicates a milestone in the manufacturing process of an item and signifies a point-in-time in the history of the life of the item.
web client	The web client provides a mechanism for users downstream of manufacturing to access information about Productstream items
worklist	A list of change orders requiring attention by a routing participant.

Autodesk Productstream

Autodesk Productstream bridges the gap between CAD data and the manufacturing process by:

- Tracking the life cycle of designs and materials used to manufacture a product.

- Managing what you make, buy, assemble, and deliver to customers.

Productstream automates the process of tracking and managing the engineering release process by giving access to the vault and its data to groups outside the scope of the engineering department. Items are managed through various working states such as work-in-progress, released, and obsolete. Bills of material (BOMs) coordinate the items in a total parts list. Change orders can be submitted, reviewed, approved, and tracked through the routing process using Productstream. Productstream Web Client is a web browser-based application that enables you to find released revisions of items as well as view and print their details.

NOTE For more information on Autodesk Productstream and all of its features, see Productstream Help.

Start Autodesk Productstream

1. Use one of these methods:
   - On the Start menu, click Programs ➤ Autodesk ➤ Autodesk Data Management ➤ Autodesk Productstream.
   - On the desktop, double-click the Autodesk Productstream icon.
2. In the Log in dialog box, enter the user name, password, the server name, and the name of the vault database.I f you do not know your user name and password, contact the vault administrator.
3. Click OK.

NOTE You can save your account information to automatically log into the vault the next time.

Working with Items

An item is a unique unit that tracks all related data to a physical part, product, or assembly. An item is identified by its item number or part number. Items typically represent:

- Parts
- Assemblies
- Consumable goods such as fluids, lubricants, and artwork

- Bills of Material

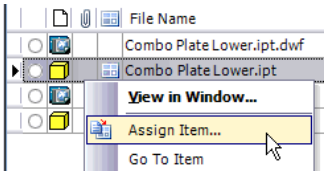
Creating Items

Items can be created in two ways:

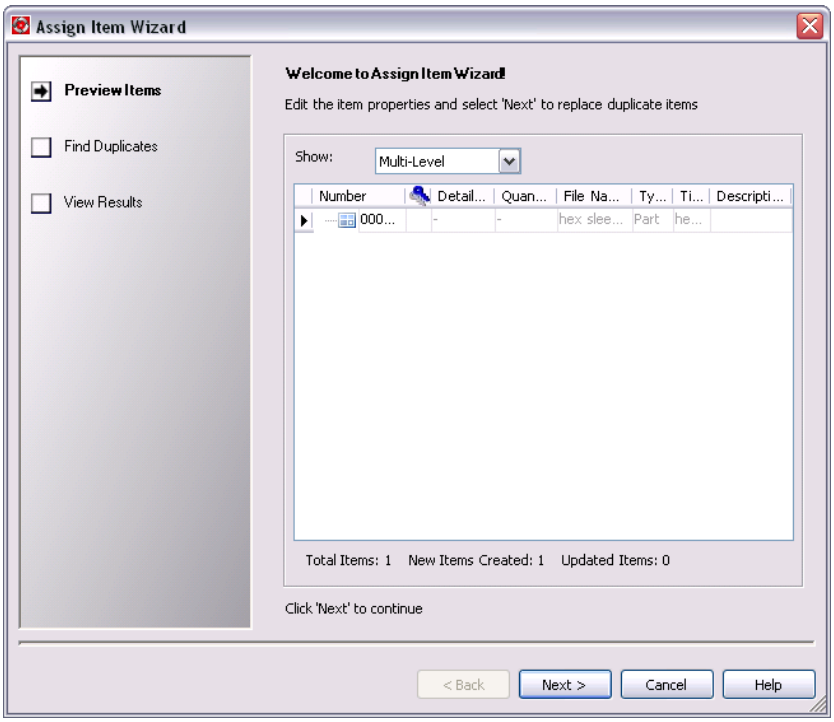
- Assign an item to a *.ipt*, *.iam*, or *.dwg* file stored in the vault using the Assign Item command. When a file is assigned an item, a corresponding item record is added to the item master. A file must be checked into the vault to be assigned an item. Each file in a vault folder can only be associated with one item. Items assigned to files contain information such as the BOM.
- Use the New Item command within the item master to create a user-defined item that is not included in the CAD design such as grease, adhesives, or mechanical components that are not modeled.

Assign an item

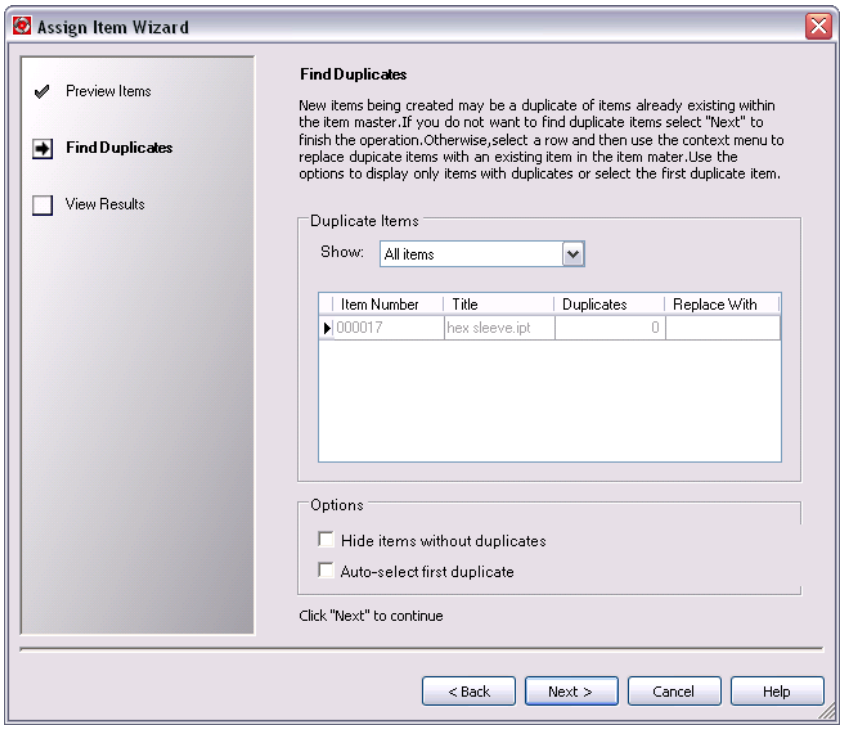
1. Select Assign Item.



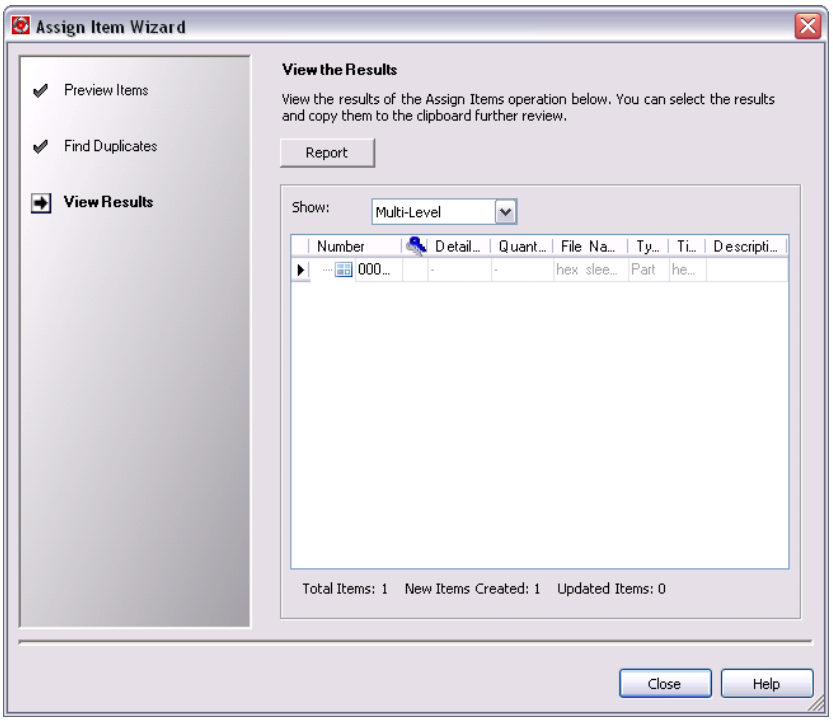
2. Preview Items and edit properties. The list of items is presented within a grid that can be changed to display the data in various formats based on the selection within the 'show' combo box. You can choose between a multi-level bill of materials view, a flat list of all items, a list of new items, or a list of updated items.



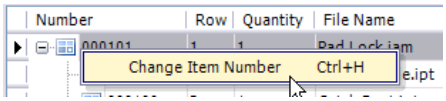
3 Scan for duplicates. You can specify an existing item to replace a new item.



4 The following dialog box is used to view the results and optionally create a report. You can select the results and copy them to the clipboard further review.

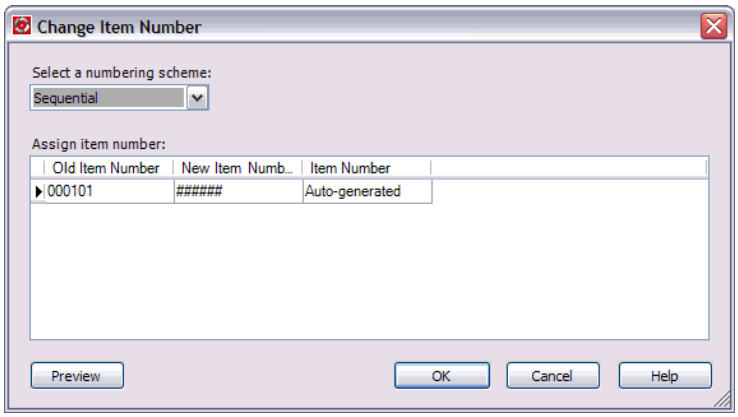


5 An item number is assigned by default. To change the item numbering, right-click an item and choose Change Item Number from the context menu.



NOTE Default numbering schemes that require user input are not applied automatically. Instead an auto-generated number is assigned. You must edit an item number requiring user input and select the item numbering scheme.

6 In the Change Item Number dialog box, select a numbering scheme from the list.



7 The preview shows all fields of the selected item numbering scheme. Fields that require user input are represented as a question mark (?) and auto-generated fields are represented as a pound sign (#).

8 Enter a number type and revision sequence.

9 Click Update to refresh Preview.

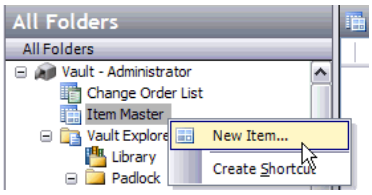
NOTE Update remains inactive until fields requiring user input are filled in.

11 Click OK to accept the new item number.

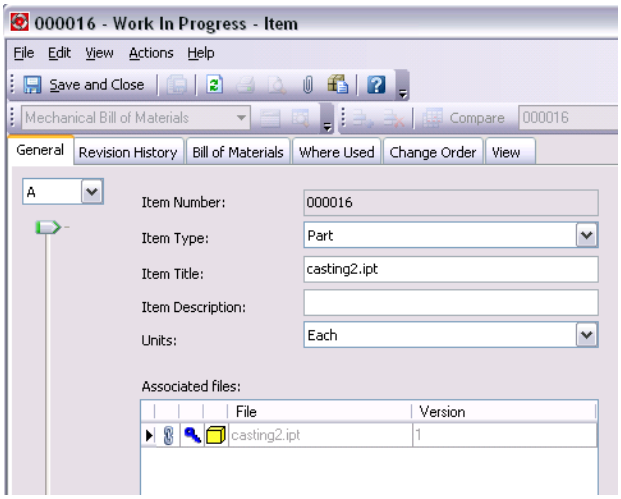
12 Click OK.

Create a user-defined item

1 Right-click Item Master in the navigation pane, and then select New Item from the context menu.



The item record is displayed and an item number is automatically generated based on the default item number format.



2 A user-defined item can be any type. From the drop-down list, select one of the following item types:

Document	The item represents any supporting documentation for the design, including spreadsheets, design specification documents, and so on.
Purchased	The item represents a part to be purchased.
Part	Indicates a part without an associated CAD file. Select when creating non modeled parts.
Assembly	Indicates grouping. For example, to create groupings of parts, such as a repair kit that is shipped with a product.
General	Groups items that do not meet another classification.
Product	The top-most assembly representing the entire product.
Process	Indicates a process, such as heat treatment, where an attached file has the specification.
Electrical	Electrical will assist AutoCAD Electrical users' out of box setup.
Electrical project	A new Electrical Project item type needs to be added to support AutoCAD Electrical bills of material

3 **NOTE** Any number of custom item types can be created. See Configure Item Types in the Autodesk Productstream Help.

4 You can perform the following operations:

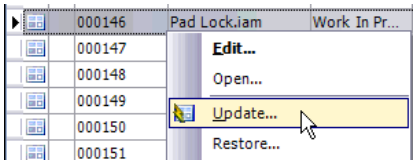
- In the Item Title text box, enter an item title.
- In the Item Description text box, enter a description.
- Click Attachments to attach a file to the item. Attachments are typically supporting files such as documents, images, and so on.
- Click Add to add a property to the item.

5 Click Save and Close to save the new item or select File ➤ Exit to close the item record without saving.

Updating Items

Update synchronizes items with the metadata in the latest file versions in the vault. Updating an item linked to an assembly file creates a new item for each new file associated with it. If the item has child items (for example, an Autodesk® Inventor™ assembly), the child items are also updated to their most recent versions. Updating an item updates BOM data from the related files in the vault.

Right-click an item in the item master, and then select Update from the context menu.



NOTE A file and its dependents must be checked into the vault to update the corresponding items.

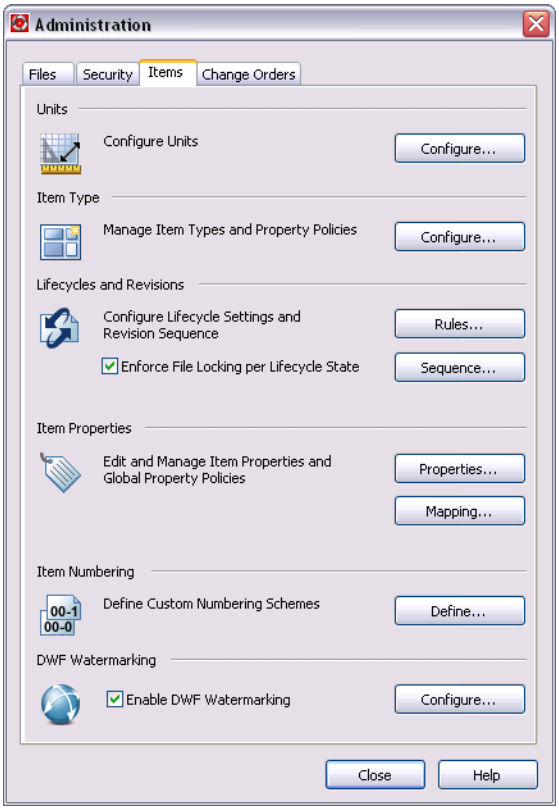
NOTE Click Update, the wizards that will appear are the same as the wizards of assigning an item except changing the name Assign Item Wizard to Update Item Wizard.

Creating Item Numbering Schemes

Autodesk Productstream comes with a default numbering scheme already in place. You can customize the numbering scheme to fit your needs. If you are using a custom numbering scheme, define the scheme before adding items. Once a numbering scheme is in use, it cannot be edited. See the Autodesk Productstream Help for more information about field types.

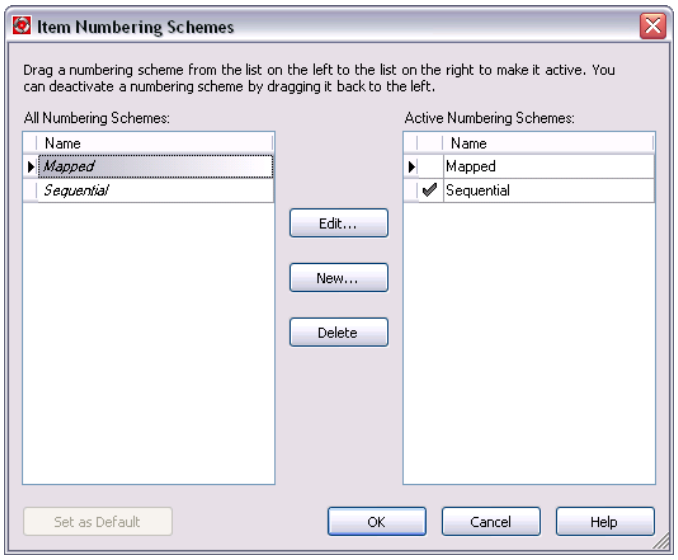
1 Click Tools ➤ Administration.

2 In the Administration dialog box, select the Items tab.

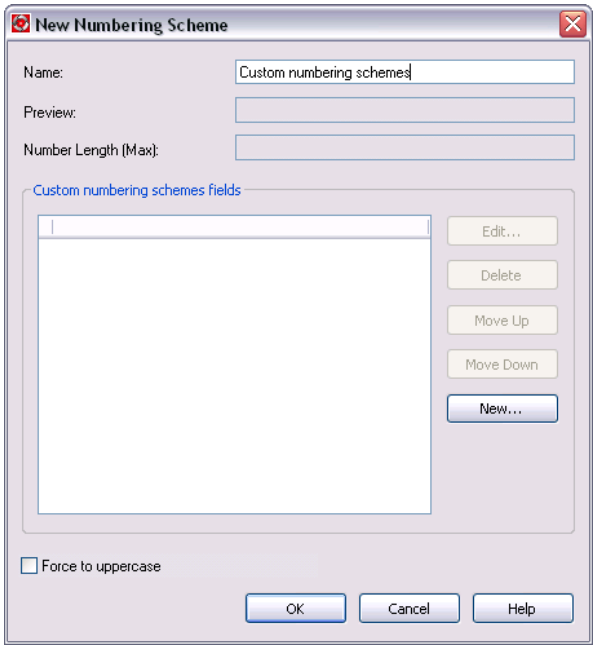


3 In the Item Numbering section, click Define.

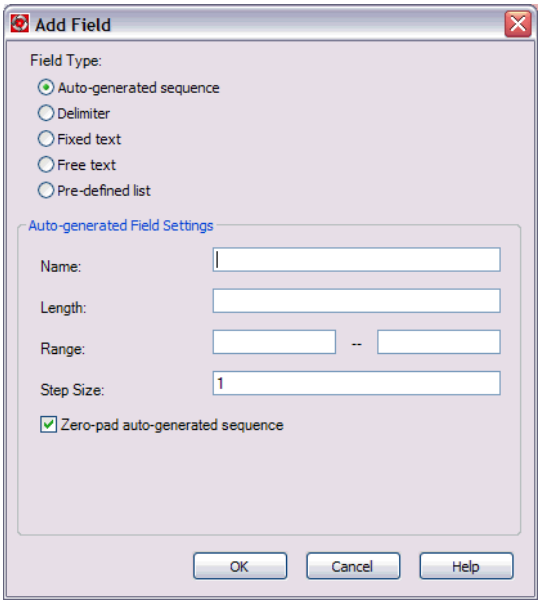
4 In the Item Numbering Schemes dialog box, click New to create a new scheme or click Edit to modify an existing one.



5 In the New Numbering Scheme dialog box, enter a name for the numbering scheme in the Name text box, and then click New.



6 In the Add Field dialog box, select one of the field types:



7 Click OK.

The New Numbering Scheme dialog box displays a preview of the settings defined in the Add/Edit dialog box.

8 To change the settings, click Edit. To delete, click Delete. To add another numbering scheme, click Add.

9 To display text in all capital letters, select the Force to uppercase check box.

10 Click OK.

Managing Item Properties

Items inherit certain properties by default when they are assigned to files. You can create custom property associations by mapping CAD properties to items and by mapping item properties to CAD data. When you map item properties to CAD data from Autodesk Inventor, you can use the Update feature in the Autodesk Inventor Add-in to write properties back to the CAD file, keeping

items and files up-to-date. See the Autodesk Productstream Help for more information on property management.

Working with Bills of Material

Every time an item is assigned to an Inventor assembly file or an AutoCAD Mechanical file that contains multiple parts, a bill of material (BOM) is generated for that item. A BOM summarizes the information needed for manufacturing a specific version of a part or product. The summary of items include a detail ID, item numbers, quantities, part descriptions, lifecycle state, and other properties that are required throughout the design release and manufacturing processes. An item representing a single part typically contains BOM information related only to that part. An item representing an assembly typically contains BOM information for all the parts in the assembly. Likewise, an item representing an assembly comprised of subassemblies and parts typically contains BOM information for all the subassemblies.

BOMs are a powerful tool for tracking a product. They can be updated as items are added and removed from the design. Custom items can be added that are outside the design process, such as grease or adhesives. You can compare one BOM revision with another to see what has changed between revisions. For example, you can pinpoint exactly which parts have changed between revisions of an assembly.

Because a BOM is an item that has other items linked to it, you can manually create a BOM by creating a user-defined item and then add items to the new item's BOM.

A BOM can be exported as a Microsoft® Excel spreadsheet, *.html* file, *.xml* file, or *.txt* file so it can be analyzed or formatted in other applications, such as Excel, for custom reports and printouts. Items can also be imported from and exported to Microsoft Business Solutions - Great Plains ERP system.

NOTE For more information on ERP integration and Great Plains, see the Autodesk Productstream Help.

Editing BOMs

Each row in a BOM represents an item. Rows can be added to and removed from the BOM as well as reordered and renumbered. While the quantities listed in the BOM are derived from the original CAD data, the quantity value

can be overridden. A BOM can only be edited in the item record. It cannot be edited in the preview pane. BOMs can only be edited at the top level. To edit a subassembly listed in a BOM, open the subassembly from the item master, and then edit the subassembly BOM.

NOTE For more information on editing BOMs, see the Autodesk Productstream Help.

Comparing BOMs

You can compare the BOM from one revision with the BOM of another revision to see what changed between revisions. The Compare tool on the BOM Edit toolbar compares the currently displayed BOM with another BOM. You can also view BOM by date and compare items' BOMs. The differences between the two revisions are highlighted in the current BOM view.

You can view effective BOM on a given date, latest BOM on a given date, previously released but currently out of date BOM on a given date.

1. In the item master, right-click an item, and then select Open from the context menu.
2. In the item record select the Bill of Materials tab.
3. Use the slider to display a BOM revision. The corresponding revision of the BOM is displayed.
4. On the BOM Edit toolbar, select a revision of the BOM or another item's BOM to compare against the displayed version.
5. Click Compare. The currently displayed BOM is compared against the revision or the BOM of another item selected in the BOM toolbar. The differences are highlighted and color coded in the BOM grid.
6. Use the navigation controls at the bottom of the BOM to step through the differences one at a time.

Working with Change Orders

Using the Change Order (ECO) dialog box, you can create change orders to describe changes to a design as well as manage the progress of that change order as it is reviewed, approved, or rejected. Change orders provide a historical record of why, how, and when changes were made.

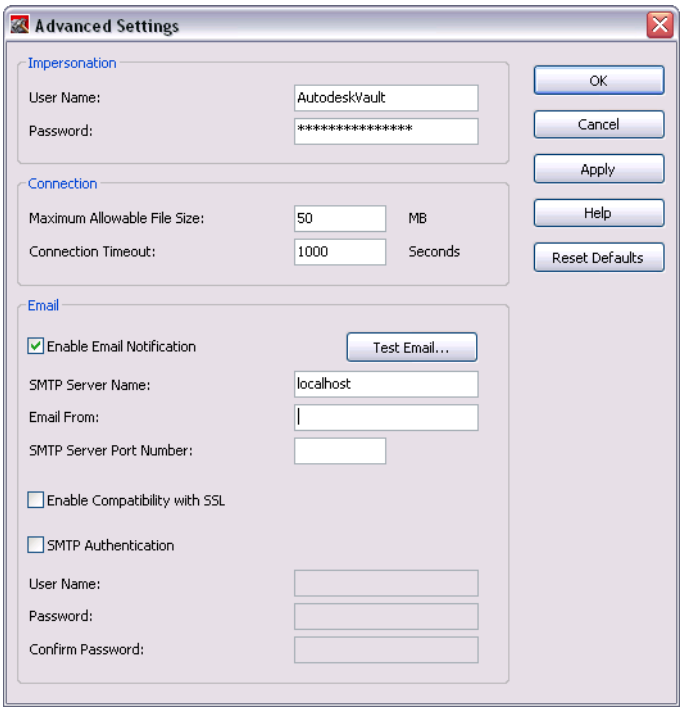
A Productstream user assigned the role of change administrator starts the ECO process by creating a change order and specifying a routing list of participants. Participants can be reviewers or approvers. Reviewers can read the change order and add comments. Approvers can read the change order, add comments, and approve or reject the change order. By default, a new user is assigned the role of Notification User. The role of Responsible Engineer has been added to the current set of workflow roles. This role gives the routing participant the ability to submit a change order for final approval. When a change administrator submits a change order, edit control over the change order is given to the Responsible Engineer(s).

- The change order number is added to each participant's worklist.
- An e-mail message is sent to each person on the routing list if they have subscribed to e-mail notification for change order events and the Autodesk. E-mail notification requires configuring the Autodesk data management server.

When a routing participant logs into Productstream, their worklist displays any change orders requiring attention and its current state. Use ADMS Console to configure e-mail notifications in Productstream for both ERP features as well as engineering change orders.

Configure e-mail notifications

1. From the Windows Start menu, select Programs ➤ Autodesk ➤ Autodesk Data Management ➤ ADMS Console.
2. Click Tools ➤ Administration.
3. In the Administration dialog box, click the Advanced Settings tab.
4. Click Email.



5 By default, the SMTP server is set to localhost. Change the value to the name of the SMTP server on the network. If you do not know the name of the SMTP server, contact your system administrator.

6 In the Email From text box, enter the from address to use for the automatically generated e-mail messages. This value is blank by default. An e-mail address must be entered. The SMTP server on the network must also be configured to send e-mail from the specified address.

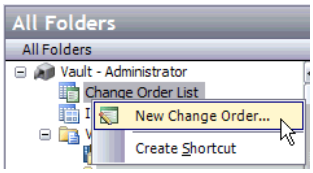
7 Enter the port number through which the SMTP server operates. This is typically port 25.

8 If the SMTP server uses SSL, turn on the Enable Compatibility with SSL check box.

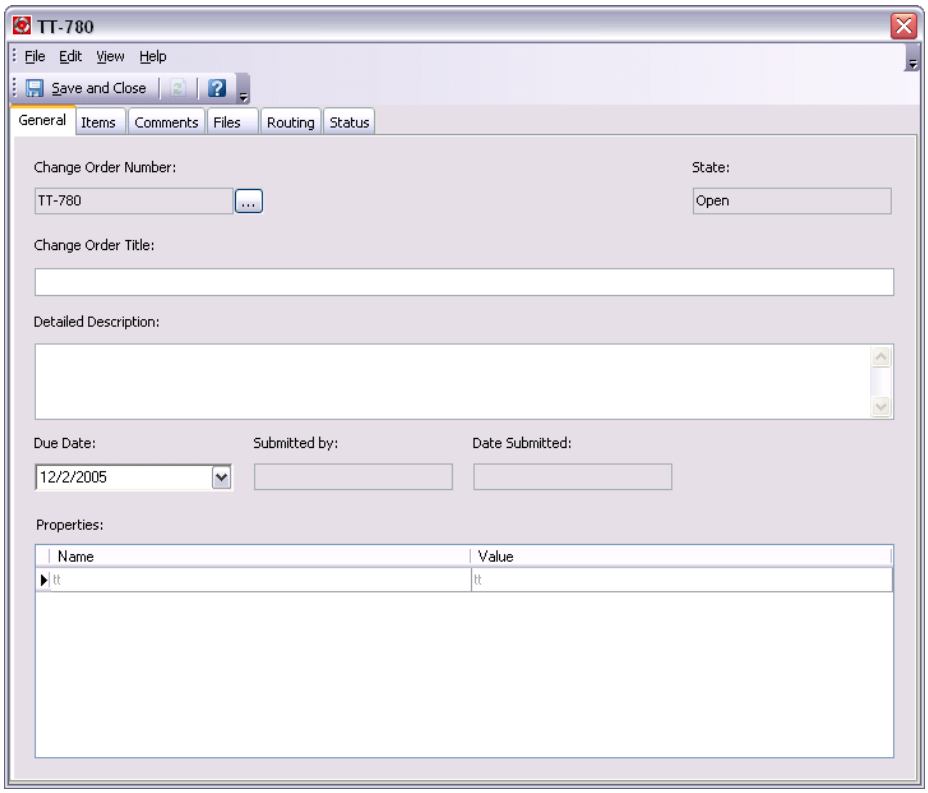
9 If the SMTP server requires authentication, turn on the SMTP Authentication check box and enter the user name and password for the SMTP account.

Create a Change Order

1 In the navigation pane, right-click Change Order List, and select New Change Order.



The ECO dialog box is displayed. A sequential ECO number is automatically assigned to the ECO. If you need to change the numbering scheme, click the browse button, and select a different numbering scheme.



2 Enter a title in the Change Order Title field.

3 Enter a description in the Detailed Description field.

4 Specify the due date the change order is to be completed by.

5 To add items, click the Items tab and select add.

6 In the Add Items dialog box, either enter search text under the Search Text field or click the Advanced tab to locate an item.

7 To attach a property to the change order, right-click the Properties field and select add.

8 Click the Routing tab, and then select a routing list.

9 To add files, click the Files tab, and then select the types of files to chose from. Select a file or multiple files, and then click Attach. You can only attach files that are in the vault.

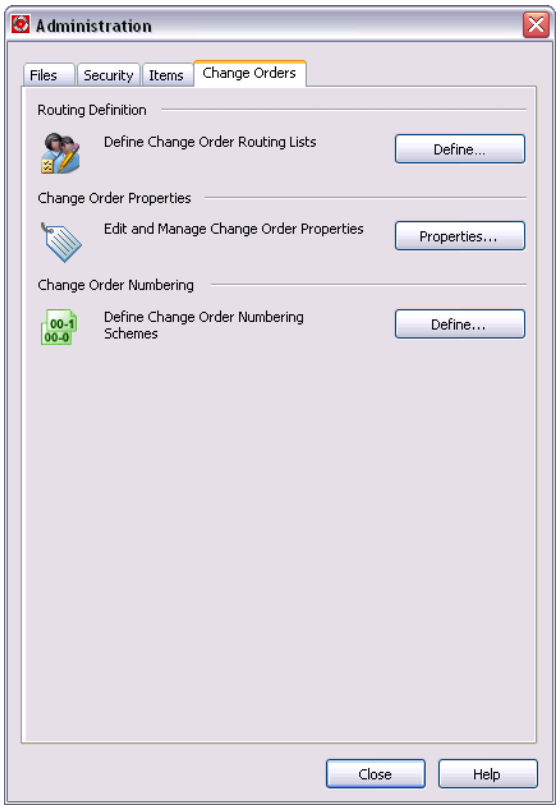
10 Click Save and Close to create the change order and exit the dialog box.

Routing lists and numbering schemes are defined in the Administration dialog box on the Change Order tab. When you first create a change list, you select from a list of predefined routing lists and numbering schemes. Be sure that your administrator has defined the appropriate numbering schemes.

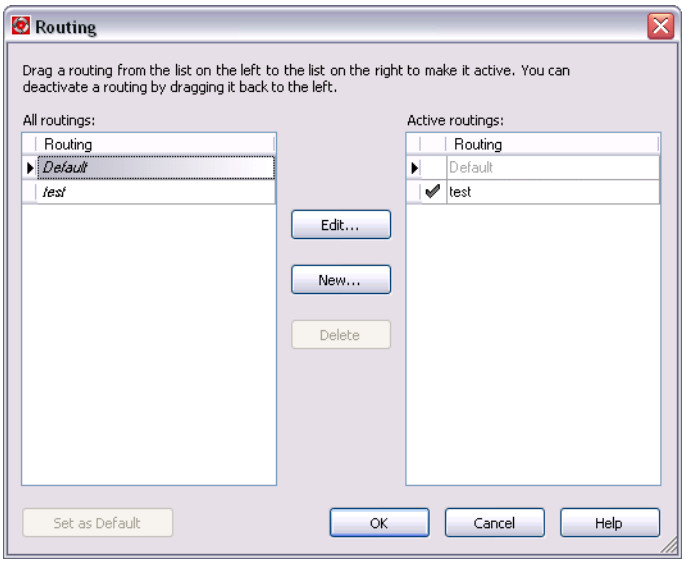
NOTE The routing list can be modified after creating the change order.

Create a Routing List

1 In the Administration dialog box, click the Change Orders tab.

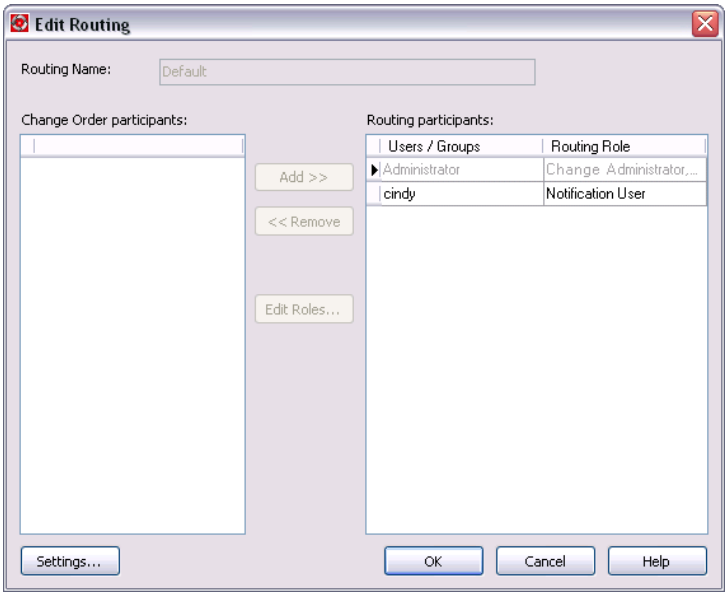


2 Click Define. The Routing dialog box displays available routings and active routings.



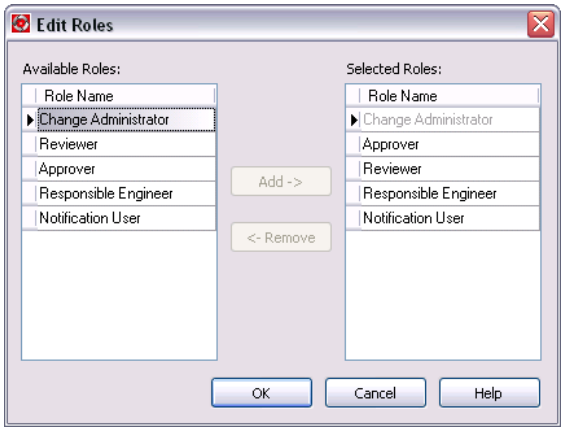
3 To create a new routing click New.

4 In the Edit Routing dialog box, enter a name for the new routing.



5 Select a name in the Change Order Participant field, and click New to add the participant to the new routing.

6 To assign one or more roles to a routing participant, click Edit Roles.



7 In the Edit Roles dialog box, select one or more roles in the Available Roles field, and then click Add to add to the Selected Role field. To remove a role, highlight a name in the Selected Role field, and click Remove.

8 If you are adding more than one approver and want to require unanimous approval, click Settings, and then select Unanimous approval required.

9 Click OK to close the dialog box.

10 Click OK in the Edit Routing dialog box to close the dialog box.

ECOs and Change Orders

ECOs help you manage the process of change in your design data, documenting participant responses through comments and markups. Approvals or rejections are tracked through the ECO system. Item effectivity can be set through a change order, specifying the date an item is to become released or expire. For more information on change orders, see Change Order Essentials in the Autodesk Productstream Help.

Autodesk Productstream Web Client

The Productstream Web Client is a Web browser-based application that enables you to find released revisions of items as well as view and print their details using Microsoft® Internet Explorer®. These include the item's bill of materials (BOM), where the item is used in other assemblies (Where Used), and attached drawings in the DWF format.

Log in and log out

1. To access the Autodesk Productstream Web Client using Internet Explorer, enter the address (URL) *http://<servername>/AutodeskDM/Webclient/default.aspx*.
2. Enter the user name and password assigned to you by the vault administrator.
3. Enter the name of the computer where the vault server is installed.
4. Enter the name of the vault database to which you have access. You can also click the Browser button to search one in the server from the Vault List in the Webpage dialog.
5. Enter the number in the Item Number field at the top of the main page.

After you have successfully logged in, the main page appears. Here you can specify items and revisions in the vault.

When you are finished working or wish to change vault servers, click the logoff button on the main view. You will log out of the current vault server and go back to the log-in page.

Item Search

If you know the item number:

1. Enter the number in the Item Number field.
2. Click the Go To button. If the item number has been released, all released revisions are displayed.

If an item number is not known: Use the search button to search for all matching items.

1. Click the Search button.
2. Enter a text string in the Search For field. All items will be searched for a match between this entry and properties of the item including title, description, and so on.
3. Click the Find button.
4. Select an item from the list and select OK. The main page appears with a list of all released revisions of this item.
5. Click Cancel to close the search dialog.

View BOMs

Click Show BOM to display the bill of materials in a new window. The tree view is displayed by default. You can expand or collapse the structure to show nested parts and assemblies. Select the Part List view to show the total parts count for the item.

NOTE This view does show intermediate subassemblies.

View Where Used

Click Show Where Used to see where the item is used in all released parent parts or assemblies.

View Attachment

If the item has an associated *.dwf*, select View Attachment to view the attached file. If the items does not have a *.dwf* attached, download the file and open it with the application associated with the file.

Configuration of Productstream

You must be assigned the role of administrator to configure the web client. Click Configure to display the configuration screen.

The Administrator can change various aspects of Productstream from the Configuration Page:

- Graphic image to be displayed in place of the standard Autodesk Graphic at the top of the web page.
- Background color.
- Columns to display for the three different forms (Item, BOM, Where Used).

The Administrator selects the Change button next to the item to be configured to make a change.

Graphic Displayed at the top of the page

This configuration option allows the administrator to replace the standard Autodesk graphic at the top of the web page. You can specify either a *.jpg* or a *.gif* file to replace the default image. See the Web Client help for more information.

Background Color

Select Change. Choose a new color for the background. You can also define and add custom colors.

NOTE Changing the background color will not change the color of the standard logo, the background color in the grid displays, or the background of the configuration page.

Customizing Columns

Select Change to display a page in which there are two columns: Available columns and Displayed fields. Modify the attributes of the form using the Add and Remove buttons.

Item Displayed

This control allows the administrator to control the display of released items in the item form. The administrator can choose between Show all released revisions and Show effective revision only.

Printing Results

1. Click Print. The print view appears in a new window. The current view of the BOM or Where Used is displayed. This view is optimized for printing using the browser's native print command.
2. From the File menus, select Print.
3. Close the Print BOM/Where Used window to return to the main view window.

Using Vaults with Microsoft Office Applications

7

Learn how to manage non design-related files in Microsoft® Word, PowerPoint®, and Excel® using the Autodesk® Vault add-in for Microsoft® Office.

Note that office applications are frequently launched in the background by other applications. For example, Microsoft® Outlook® starts Word in the background if you use Word as your e-mail editor/viewer; Autodesk® Inventor™ launches Excel in many workflows such as parameters, iParts, and so on.

Key Terms

Term	Definition
check in	Returning a modified file to the vault. The previous version is retained in the file history.
check out	Retrieving a read/write copy of a file stored in the vault for editing.
local copy	The copy of the vault source file placed in the local working directory.
working folder	A local folder to which files are copied from the vault for editing. A working folder must be mapped to the root $.

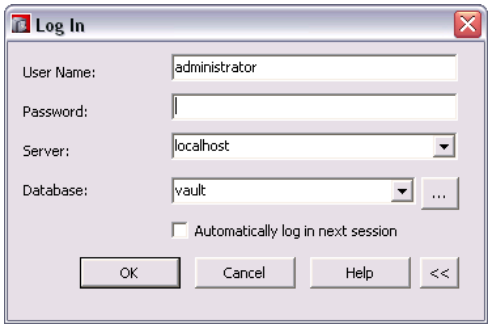
Integrating with Microsoft Office Word

To manage files that are stored in the vault using the Vault Add-In for Microsoft Office, you must log into the vault database. If an account has not been set up for you, contact the vault administrator.

For more information about using the Vault add-in for Microsoft Office, see Autodesk® Vault Help available from the Help menu in each of the Microsoft Office applications.

Log into a vault

1. Open a Microsoft Office application.
2. Click the Login button in the toolbar or click File ➤ Autodesk Vault ➤ Log In.



3. Enter the user name and password assigned to you by the vault administrator
4. Enter the name of the computer where the vault server is installed. If you do not know the name of the vault server, contact the vault administrator.
5. Enter the name of the vault database to which you have access. You can select the name of the database from the list or click the expansion button to see the complete list of databases.
6. Click OK.

Log out of a vault

1. Click File ➤ Autodesk Vault ➤ Log Out. The Log out was successful dialog box is displayed indicating that you are no longer logged into the vault

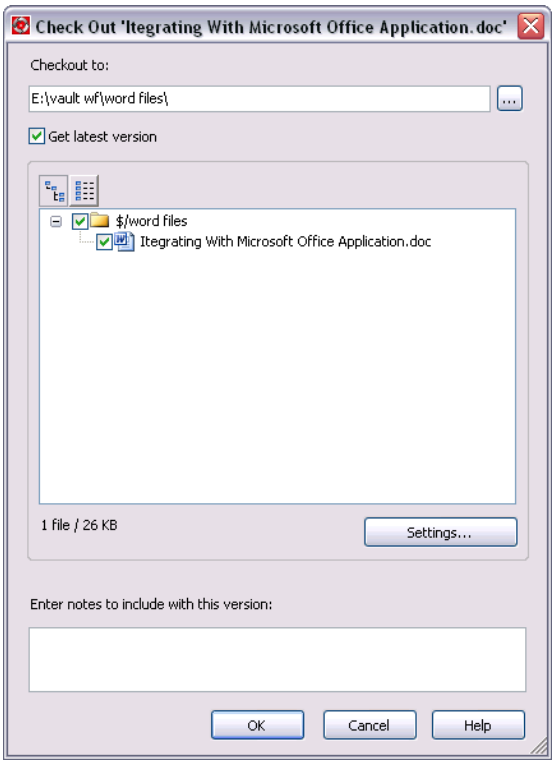
2 Click Close to completely exit the vault, or click Log In to log back into the vault.

Checking Out Files

To make changes to a file, you must first check it out of the vault. No one else can check out a file that is checked out until it is checked back in to the vault. You can check out any number of files.

1 Select File ➤ Autodesk Vault ➤ Check Out.

2 In the Check Out dialog box, enter a comment and specify whether or not to get the latest version.



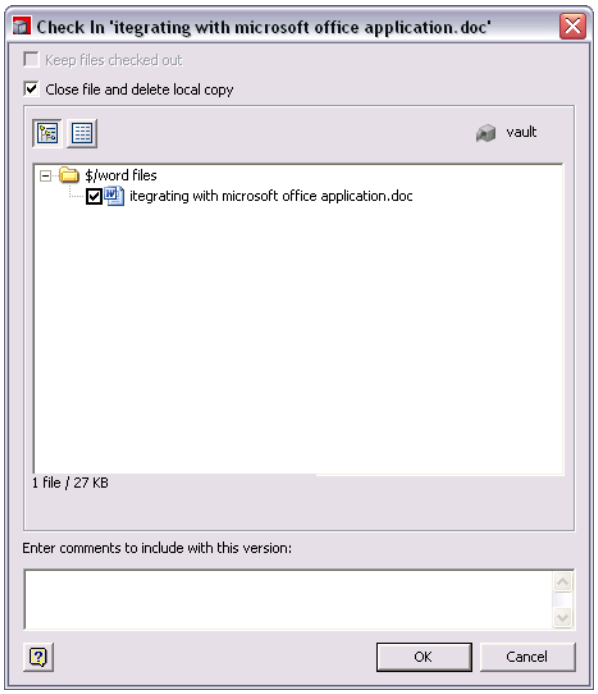
3 Click OK.

Checking In Files

As you work on Microsoft Office files, you can save changes to the local copy. To record the changes in the vault, you must check the file in.

Check in a file

1. Save the file. If the file is not saved, you are prompted to save the changes.
2. Click File ➤ Autodesk Vault ➤ Check In.
3. In the Check In dialog box, enter a comment and specify whether you want to keep the file checked out for further editing or close the file and delete the local copy.



4. Click OK.

Undoing Check Outs

Cancel changes to the selected files both in the vault and in the local folder. You are given the option to retrieve the latest version of the file from the vault overwriting any changes you made to the document locally.

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